Rev	ECO #	Date	Revision Description
J	0061233406	29 Nov 2024	Revised multiple sections

Flexible Printed Circuit Board Specifications

080-2265

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Table Of Contents

1. DOCUMENT DESCRIPTION....6
   1.1. Scope....6
   1.2. Purpose....6
   1.3. Referenced Documents....6
   1.4. Classification Requirements....8
   1.5. Terms And Definitions....8
   1.6. Precedence of Documents....8
   1.7. Responsibilities....9
   1.8. Notification....9
   1.9. Modifications....11
   1.10. Regulatory Requirements....11
   1.11. Workmanship....12
2. MATERIAL....13
   2.1. Material Details....13
   2.2. Dielectric Materials....13
   2.3. Multilayer Stack-up....14
   2.4. Dielectric Constant and Loss Tangent....14
   2.5. Surface Imperfections....14
   2.6. Copper Foil....14
   2.7. Halogen-free Materials....14
   2.8. Environmental Specifications....14
3. DRILLED HOLES....15
   3.1. Hole Location Tolerance....15
   3.2. Hole Diameter....17
   3.3. Minimum Annular Ring requirement....18
   3.4. HDI Stack-up Requirements....21
4. COPPER PLATING....22
   4.1. Copper Plating Quality and Integrity....22
   4.2. Plated Copper purity....22
   4.3. Plated Copper Thickness....22
5. CONDUCTIVE PATTERNS....24
   5.1. Pattern Location....24
   5.2. Width....24
   5.3. Impedance Control....25
   5.4. Repairs....26
   5.5. Separation....26
   5.6. Trace Tapers/Tear Drops....27
6. PHOTO-IMAGEABLE SOLDERMASK AND COVERLAY....28
   6.1. Materials and Application....28
   6.2. Photo Imageable Soldermask Thickness....28
   6.3. Photo Imageable Soldermask Registration....28
   6.4. Photo Imageable Solder mask coverage....30
   6.5. Coverlay / LPI Registration opening size and Adhesive Squeeze out....31
   6.6. Photo Imageable Solder mask and Coverlay Overlap....33
7. SURFACE FINISHING / COATING....34
   7.1. Material thickness and Application type....34
   7.2. Electroless Nickel/Immersion Gold....34

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7.3. Electrolytic Gold / Nickel Plating 34
7.4. Organic Solderability Preservative (OSP) 34
7.5. Electroless Nickel/Electroless Palladium/Immersion Gold (ENEPIG) 35
7.6. Direct Electrolytic Gold 35

8. HOT BAR PAD DESIGN 36
8.1. Material selection 36
8.2. Dimension specification 36
8.3. Hot Bar Pad Solder Mask Thickness 40
8.4. Pre-soldered Thickness (After Reflow) 43
8.5. Dimensional Stability and pre conditioning test 44
8.6. Fiducial Transparency 45

9. MECHANICAL AND DIMENSIONAL REQUIREMENTS 46
9.1. FPCB Thickness 46
9.2. FPCB Outline 49
9.3. Working panel or shipping array dummy areas 50

10. STIFFENER APPLICATION 51
10.1. Material 51
10.2. Visual inspection 51
10.3. Stiffener Bond Strength to FPC 51

11. LEGEND MARKING 52
11.1. Application and Information 52
11.2. Legend Ink color 53
11.3. Accuracy 53
11.4. Durability 53

12. CLEANLINESS 54
12.1. Cleanliness 54
12.2. In-Process Cleanliness 54
12.3. Ionic Contamination 54
12.4. Wire bonding pad Cosmetic and Cleaning 54

13. ELECTRICAL REQUIREMENTS 55
13.1. Electrical Test 55
13.2. Impedance Test 55

14. QUALITY REQUIREMENTS 56
14.1. Deposition 56
14.2. Visual Inspection 56
14.3. OK2Ship Report 56
14.4. In Process Inspection 56
14.5. Reports 57
14.6. Moisture 57
14.7. Compliance to Industry Standards 57

15. ENVIRONMENTAL TEST METHODS 58
15.1. Continue Quality Reliability Assurance 58

16. STORAGE AND PACKAGING 59
16.1. Cleanliness and Moisture 59
16.2. Storage Conditions 59
16.3. Packaging 59
16.4. Shelf Life 60
16.5. Retained samples 60

17. SHIPPING REQUIREMENTS 61
17.1. Shipping Requirements 61

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APPENDIX A: REVISION HISTORY 62

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1. Document Description

1.1. Scope

This specification applies to flexible printed circuit boards (FPC) produced by chemically etching conductor patterns on insulating base material with interfacial connections established by mechanical or laser drilled holes in the insulating base material. The Flexible Printed Circuit Boards refer to single sided, double sided and multilayer with or without stiffener, EMI shielding and PSA.

1.2. Purpose

The purpose of this specification is to establish the requirements for single sided, double sided and multilayer flexible printed circuit boards procured as a substrate or as an assembly for Apple Inc. products. This document also establishes some required practices for quality control, reporting and record keeping by the fabrication suppliers of FPC to be used in Apple, Inc. products.

1.3. Referenced Documents

The latest revisions of specifications noted below will be applicable.

General documents

IPC-A-610 Acceptability of Electronic Assemblies
IPC-T-50 Terms and Definitions for Interconnecting and Packaging Electronic Circuits
IPC 2615 Printed Board Dimensions and Tolerances

Design specifications

IPC-2223 Sectional Design Standard for Flexible Printed Boards

Material specifications

AMS 2422 Gold Plating
AMS 2403 Nickel Plating
ASTM B488 Electrodeposited Coatings of Gold for Engineering Uses
IPC-4202 Flexible Base Dielectrics for Use in Flexible Printed Circuitry
IPC-4203 Adhesive Coated Dielectric Films for Use as Cover Sheets for Flexible Printed Circuitry and Flexible Adhesive Bonding Films
IPC-4204 Flexible Metal-Clad Dielectrics for Use in Fabrication of Flexible Printed Circuitry
IPC-FC-234 Pressure Sensitive Adhesives Assembly Guidelines for Singles and Double sided Flexible Printed Circuits

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Performance and inspection documents

IPC-A-600 Acceptability of Printed Boards
IPC-6013 Qualification and Performance Specification for Flexible Printed Boards
IPC-6202 IPC/JPCA Performance Guide Manual for Single- and Double-Sided Flexible Printed Wiring Boards
JPCA-DG02 Performance Guide for Single and Double-sided Flexible Printed Wiring Boards
IPC 7711/21 Rework, Modification and Repair of Electronic Assemblies

Flex assembly and materials standards

IPC-FA-251 Guidelines for Assembly of Single- and Double-Sided Flex Circuits
IPC-FC-234 PSA Assembly Guidelines for Single- & Double-Sided Flexible Printed Circuits
IPC-3406 Guidelines for Electrically Conductive Surface Mount Adhesives
IPC-3408 General Requirements for Anisotropically Conductive Adhesives Films

Apple Document
080-01409 Apple Bare FPC Critical Process Parameters
080-03022 Apple Bare FPC Quality Control Plan
080-03267 Apple FPC Incoming and outgoing Cosmetic Criteria
080-02864 Bare FPC shelf life Extension
080-02999 Apple Bare FPC General Continuous Quality Reliability Assurance
080-02882 NPI OK2Ship Report for Bare FPC
080-02881 MP OK2Ship Report for Bare FPC

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1.4. Classification Requirements

For any characteristic not specified in this or other referenced documents, the default specification shall be defined in IPC-6011, IPC-6013 (latest revision), IPC-6202, IPC-A-600 and IPC-A-610 (latest revision). In such case, Apple FPCB's shall confirm to class 2 requirements or based on application specified by Apple documents and authorized Apple DFM/SQE.

1.5. Terms And Definitions

Definitions used in this specification are in accordance with IPC-T-50.
Additional terms and their definitions are:

DFM	Design For Manufacture
FPC	Flexible Printed Circuit Board
FPCA	Flexible Printed Circuit Board Assembly
Top/Bottom Side	The two external surfaces of a FPCB/FPCBA
PTH	Plated Through Holes
NPTH	Non Plated Through Holes
HDI	High Density Interconnection
ENIG	Electroless Nickel Immersion Gold
ENEPIG	Electroless Nickel Electroless Palladium Immersion Gold

1.6. Precedence of Documents

In case a conflict exists between documents, the following precedence of documents shall apply:

- Purchase Order
- ODB++ Data, 1-up FPCB
- ODB++ Data, panel array, if applicable
- MCO drawing
- README File
- ERS File
- Apple P/N: 080-2265 (this document), current revision
- IPC-6011 and IPC-6013, (latest revision)
- IPC-A-600 and IPC-A-610 (latest revision)
- IPC-6202/ JPCA -DG02-3ed (latest revision)

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1.7. Responsibilities

Apple, Inc. will be responsible for the content of the ODB++ data and associated deliverables made available for the fabrication of FPCB's for Apple products. If Gerber data is provided by Apple, it is for reference only and may not be used for fabrication of the FPCB. Only ODB++ design data is to be used for FPCB fabrication.

The FPCB fabrication supplier is responsible to verify that the data, including all documentation obtained from Apple for the fabrication of FPCB's is the same revision level as that specified by the FPCB purchase order. The FPCB fabrication supplier will also be responsible to verify the instructions/data received from Apple are proper for the fabrication of the FPCB as requested by Apple, Inc.

The FPCB fabrication supplier will be responsible for the fabrication of FPCB's in compliance with this specification unless an authorized Apple representative has provided deviations from these requirements in writing.

It is the fabrication supplier's responsibility to target design nominal for all features:

- The actual **average** value of laser-drilled-PTH size (but not BVH and IVH size), solder mask opening size, metal pad size, and trace width shall meet **±5 μm or ±10% of target design** nominal value whichever is less. Design tolerances are intended **only** to **accommodate for process variations**. It is not acceptable to change one feature to enhance another. For example, it is not acceptable to target a lower copper thickness to make it easier to etch fine traces.
- Controlled Impedance design nominal is the target impedance. The trace width may be modified to meet the impedance requirement, and must meet specification. Refer to section 5.3. Impedance Control.
- **Nominal** of an unbalanced tolerance is defined as the **target value**. For example, 2.0 is the target value in the specification 2.0 +0.05 -0.08 mm.

1.8. Notification

Potential deviations or modifications from Apple specification or design data, as noted in this document, must be approved in writing by an authorized Apple representative prior to the initiation of FPCB fabrication by the supplier.

The FPCB fabrication supplier shall contact the designated Apple representative at dfflex@group.apple.com and pcb-verification@group.apple.com with regard to any recommendations for improvements to design, layer structure, etc. Apple requires a complete DRC/DFM report and stack-up diagram including dielectric constant and all material thicknesses, including copper, be provided for Apple review at dfflex@group.apple.com and pcb-verification@group.apple.com.

The report should follow and contain a minimum of :

- Use Apple Engineering Deviation Request form 080-5473, Revision 02

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- Follow Apple File naming rule for correspondence … Flex DFM <project number> < application type > < build level > < part number > < FPCB fabrication supplier > <Date>
- Detail material stack up with different layer section and before and after board thickness / tolerance. Along with stack up, FPCB fabrication supplier must disclose material part number with type of raw material used, in the case of more than one material vendor are selected all applicable part number shall be included in stack up.
- Full MCO mark up with any discrepancy issue found.
- A description (preferably a picture) of any feature suggested to be changed.
- Detail (preferably a picture) of how the applied change should appear.
- A reason for the suggestion and detail on improvement to yield, productivity and cost.
- Priority Colour coding if any

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1.9. Modifications

Modifications to the ODB++ design data provided by Apple must be approved in writing by an authorized Apple representative unless the modification will continue to satisfy all of the specification limits for the ODB++ data or unless they are among the following list of acceptable modifications to Apple ODB++ data:

- Scaling – Incorporate scaling factors to compensate for material shrinkage/ expansion in the FPCB fabrication process
- Artwork – Modification for smooth traces and trace taper / tear drop. FPCB fabrication supplier must perform design rule check to verify no violation. See detail artwork modification specification in section 5 .

The FPCB fabrication supplier shall submit plots of the final manufacturing design data with approved edits to Apple.
Any material change must highlighted specifically and got approved from Apple authorized person or they are non-approved deviation.

1.10. Regulatory Requirements

All FPC shall be in compliance with UL 796 and the UL file (summarising FPC stack up and UL logo for each project) shall be in accordance with that document as well as the FPCB design database. The UL file shall include the following elements:

- UL/cUL recognition mark for US and Canada
- Manufacturer's UL file number
- FPC material designation as certified by UL, both:
  - 105°C MOT
  - Flammability rating, minimum UL 94 V-0 or VTM-0 unless otherwise specified in the README file or ODB++ data.

The FPCB fabrication supplier is responsible for the maintenance of UL certification and must show proof of UL compliance as well as material traceability, to Apple DRI team, including EPM and Safety team DRI, for all sustaining and new projects APN.

For all new projects APN:

- **Application**: as soon as FPCB fabrication supplier receives MCO or ODB++ for the first time (regardless of the build name, POC, P0, P1, P2, C6, C5, etc.), FPCB Fabrication supplier shall immediately apply for UL certification, and communicate timeline to Apple DFM team.
- **Certification**: FPCB fabrication supplier shall get the UL approval / UL certification / UL file number **before EVT**. If not feasible, FPCB fabrication supplier shall highlight to Apple DFM team, and provide schedule until completion.

The FPCB fabrication process shall comply with P/N 069-0135, Apple Specification, and Regulated Substances.

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1.11. Workmanship

FPCB fabrication suppliers shall fabricate FPC for Apple in such manner as to be uniform and consistent in quality as defined by this specification and shall be free from contaminants including oils, salts, silicones and all foreign materials which may affect the manufacturability, shelf and/or service life, serviceability, and/or appearance of the FPC.

All dimensional requirements stipulated in this specification shall apply to the FPC as delivered, as well as at the completion of standard assembly operations, as applicable.

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2. Material

2.1.Material Details

The stack-up detail showing layer-to-layer construction for FPCB's will be specified in the drill drawing layer of the ODB++ data.

Additional detail, which specifies localized thickness requirements and/or the presence or absence of specific dielectric layers will be present in the MCO.

2.2.Dielectric Materials

All dielectric material shall conform to UL designation with a flammability rating of UL 94V-0 or UL 94 VTM-0 as applicable. All dielectric material shall also conform to IPC-4202, IPC-4203, IPC-4204 as applicable.

For dielectric materials (except adhesives) shall have a minimum glass transition temperature minimum (Tg) of 140°C as measured in accordance with IPC-TM-650 2.4.24.1 and a minimum decomposition temperature (Td) of 320°C as measured in accordance with IPC-TM-650 2.4.24.6. Adhesives shall be selected as appropriate for the FPC based on application and must at a minimum withstand the reliability requirements as outlined in this document.

Additional dielectric base materials, necessary to meet electrical, thermal and/or mechanical requirements, must either be specified in the ODB++ data or approved in writing by an authorized Apple representative prior to inclusion in the multilayer stack up of an Apple FPCB.

The insulation resistance of the Flexible Copper Clad Laminates (FCCL), bonding sheet, coverlay and Solder Mask shall be greater than 100 MΩ in accordance with IPC-6013 - 3.9.4 requirement and test condition refer to IPC-TM 650 - 2.6.3

The peel strength of the copper clad laminate, as received by the FPCB fabrication Supplier, shall be greater than or equal to six pounds/in (1.05 N/mm) when tested in accordance with IPC-TM-650-2.4.9. Any deviation to the peel strength specified requires approval from Apple authorized person. This peel strength requirement does not apply to materials specified in ODB++, MCO, and ERS, which have been selected intentionally.

A proper copper adhesion treatment and multilayer lamination process shall be employed such that the resulting inner layer bond strength is greater than or equal to four pounds/in when tested in accordance with IPC-TM-650-2.4.40.

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2.3.Multilayer Stack-up

- The individual copper and dielectric thickness requirements specified on the drill drawing layer of the ODB++ data are intended to reflect requirements for finished FPC.
- Multilayer stack-ups must be designed to 1) Minimize bending stresses, 2) Optimize dimensional stability, and 3) Optimize FPCB costs and material availability.
- The FPCB fabrication supplier shall provide a complete stack-up accordingly with different layer zone including dielectric constant and all finished material thicknesses, including copper foil, solder mask, coverlay, adhesive, EMI shielding, Stiffener, PSA, Surface finishing type and plated copper. Along with material vendor name and part number the type of copper foil and type of adhesive shall be disclosed, as well as whether a button-via plating process, selective plating process or a panel plated copper process is used.

2.4.Dielectric Constant and Loss Tangent

The dielectric constant of the dielectric material shall be 2.9 to 3.6 when measured at 1 GHz and the loss tangent shall be maximum 0.040 when measured at the same. When black coverlay is required the dielectric constant of the dielectric material shall be less than 5.2 when measured at 1 GHz and the loss tangent shall be maximum 0.100 when measured at the same. Any deviation to the specification require approval from Apple authorized person.

2.5.Surface Imperfections

FPCB surface shall be inspected in accordance with 080-03267 Apple FPC Cosmetic Spec and/or dedicated project cosmetic spec provided by Apple authorized person. For any items non specified in that document, FPCB fabrication supplier shall refer to IPC-6013 and shall confirm class 2 requirements, or specifications based on application from Apple documents and Apple authorized DFM/SQE.

2.6.Copper Foil

HTE copper foil shall be used for Apple FPC unless otherwise specified on the stack-up detail of the drill drawing layer in the ODB++ data or the MCO.

2.7.Halogen-free Materials

“Halogen-free” refers to FPC laminate; solder mask, coverlay, legend ink, and surface finish, which shall comply with Apple P/N 069-1857, Apple Halogen-Free Specification. Halogen-free materials shall comply with all other Apple material specifications.

2.8.Environmental Specifications

All Apple FPCB’s must comply with the following Apple specifications:

- P/N 069-1111, Apple RoHS Compliance Specification
- P/N 069-0135, Apple Regulated Substances Specification

Finished Apple FPCB’s shall not contain Triphenyl phosphate (TPP) or red phosphorus.

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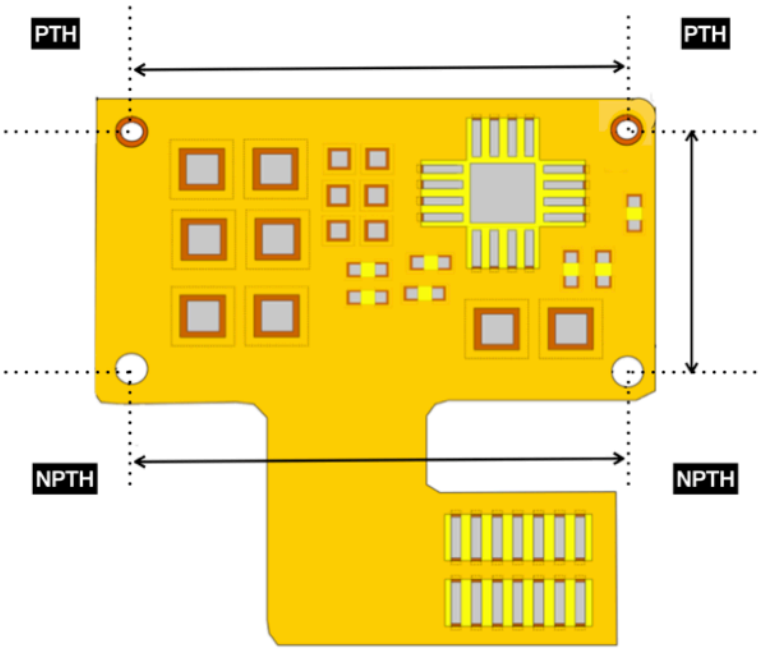
3. Drilled Holes

3.1.Hole Location Tolerance

The individual hole location requirements specified on the drill drawing layer of the ODB++ data are intended to reflect requirements for finished goods.

NPTH refers to Non plated through hole
PTH refers to Cu plated through hole.

Below picture show the definition of NPTH and PTH in finished cavity FPC



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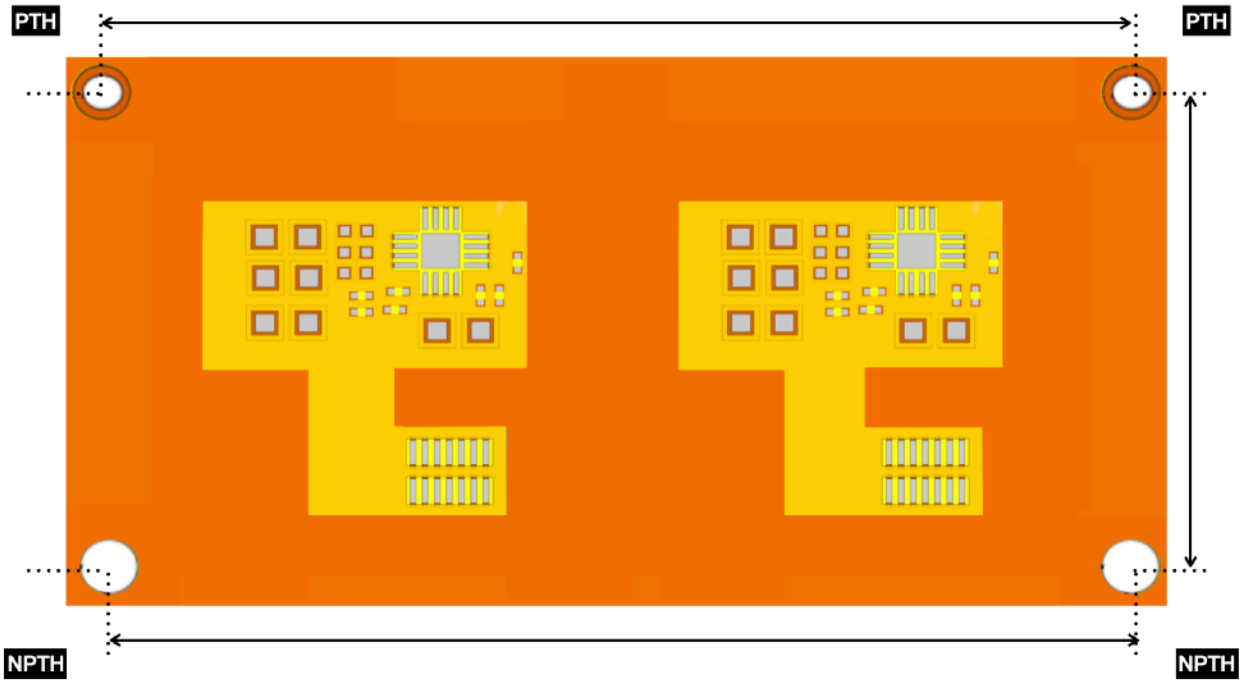
Below picture show the definition of NPTH and PTH in FPC panel



The PTH and NPTH can be formed by mechanical punch hole (by die), mechanical drilling (by NC drill bits), target punch (puncher) and Laser drilling.

Unless specified by MCO, below table show accuracy tolerance requirement exclude via for layer pattern connection:

Hole type	Position accuracy tolerance (Finished FPC size with MD or TD ≤ 200 mm)	Position accuracy tolerance (finished FPC size with MD or TD > 200 mm)
NPTH to NPTH	Nominal ±75 μm	Nominal ±100 μm
NPTH to PTH	Nominal ±100 μm	Nominal ±200 μm
PTH to PTH	Nominal ±75 μm	Nominal ±150 μm

Any deviation to the specifications require approval from Apple authorized person.

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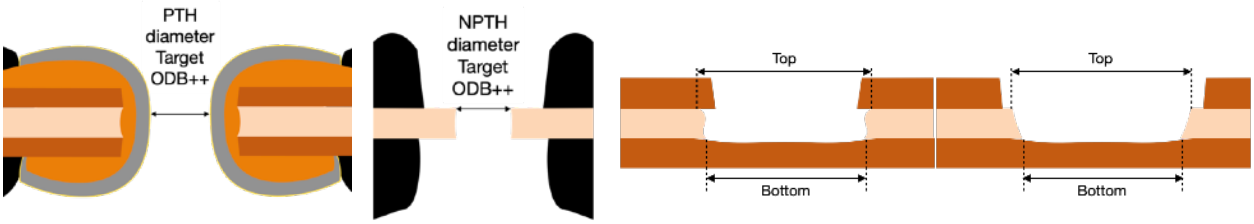
3.2. Hole Diameter

Hole diameter **target** is specified in ODB++ or MCO, based on below **definition**. Refer to illustrations for measurement locations.

- **PTH** (including through via for hot bar) **diameter**: **metal to metal** at **Finished Goods level**, narrowest location, usually measured from top with light from bottom. For example, 0.1 mm nominal in ODB++ means FPCB fabrication supplier shall drill larger and then plate to meet 0.1 mm nominal at Finished Goods level.
- **NPTH diameter**: dielectric to dielectric, top diameter, usually measured from top with light from bottom.
- **BVH diameter**: dielectric to dielectric, top diameter.



PTH diameter | NPTH diameter | BVH diameter, overhang shape (e.g. direct laser method) | BVH diameter, opening shape (e.g. conformal method)

- Hole diameter **tolerance** should follow ODB++ or MCO or below table 3.2, whichever is tighter.

Table 3.2. Hole diameter tolerances

Forming method	PTH (includes through via for hot bar)	NPTH	BVH
Target punch (Puncher)	±80 μm	±50 μm	N/A
Mechanical drilling (NC drill bits)	±80 μm	±50 μm	N/A
Mechanical punch (die)	±80 μm	±80 μm	N/A
Laser (UV or CO_2)	**After drilling**: ±15 μm di-electric to di-electric, with FPC supplier internal target **FG level**: refer to MCO, ODB++ for target and tolerance	±15 μm	±15 μm

Any deviation requires written approval from Apple authorized person.

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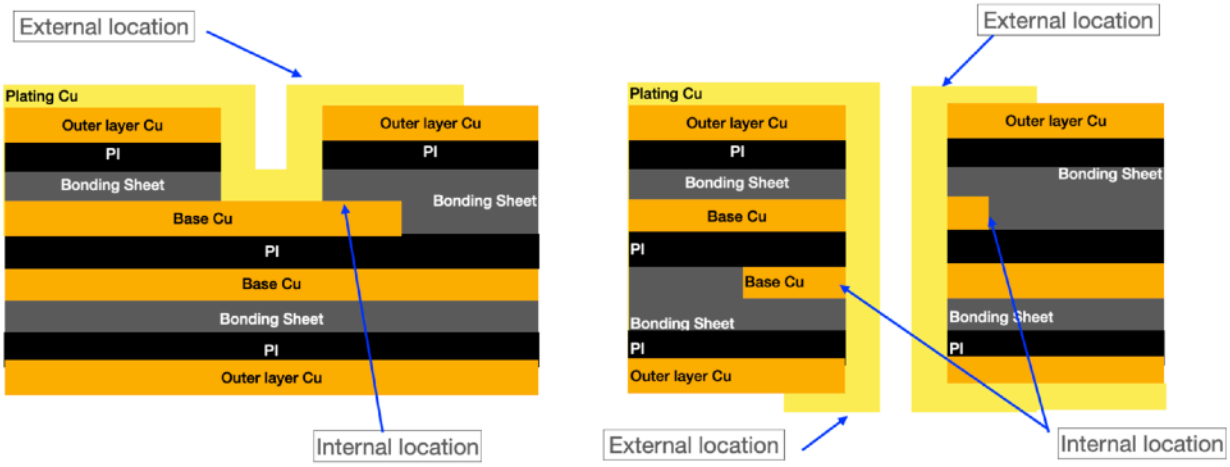
3.3.Minimum Annular Ring requirement

There is 2 types of copper plating: panel plating and button plating. Below pictures show the images of panel plating and button plating, via and pattern pad.

External and internal annular ring location show as below picture



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3.3.1 Minimum Annular Ring (External, via top) for Panel Plating

The PTH/ via minimum annular ring shall be 12.5μm for external layers inclusive of plating.

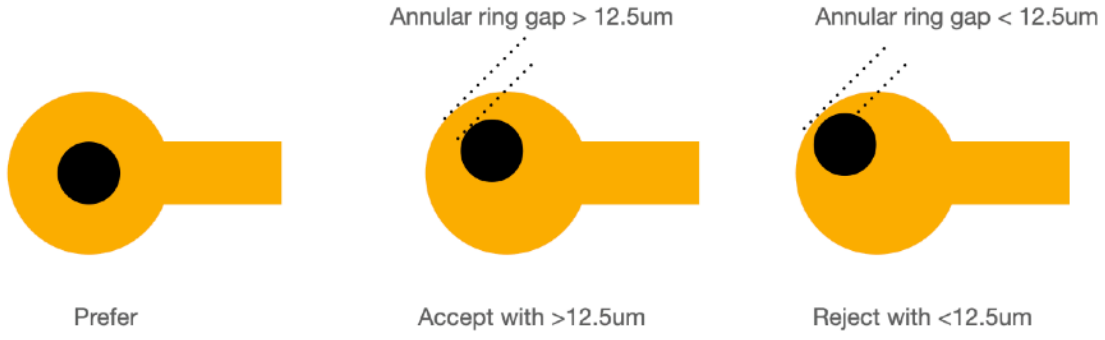

Figure 3.3.1. Via Breakout Allowances, External Layers

3.3.2 Minimum Annular Ring (Internal, via bottom) for Panel Plating and button plating

The PTH/ via minimum annular ring for internal layers shall be 0μm exclusive of plating except for the trace to pad junction, which shall have a minimum 25μm annular ring exclusive of plating. Hole to pad breakout is not allowed.

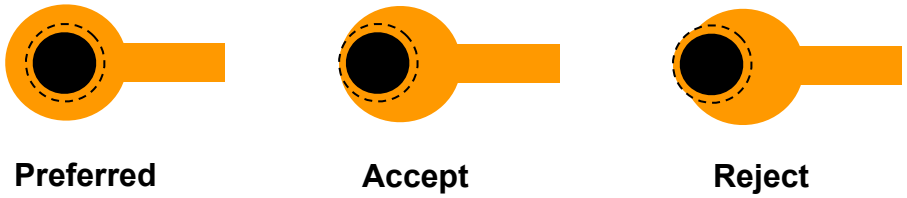

Figure 3.3.2. Via Breakout Allowances, Internal Layers

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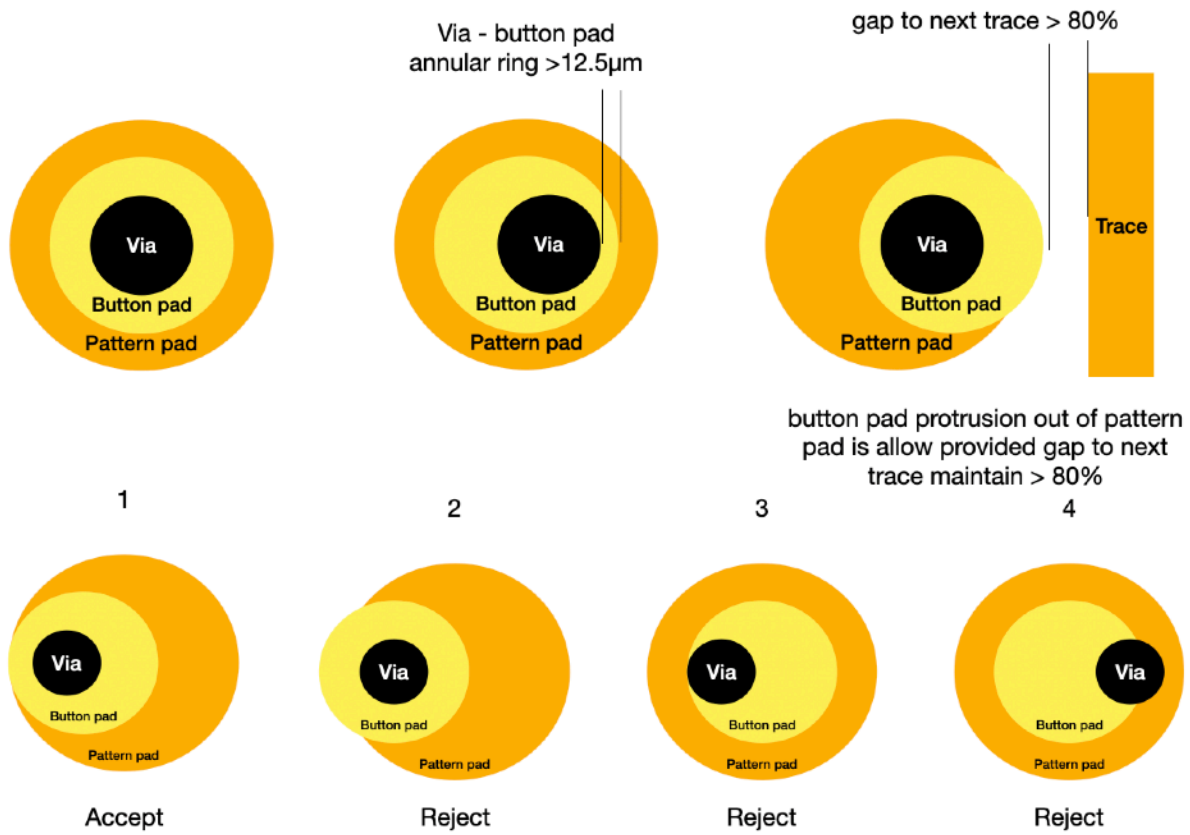
3.3.3 Minimum Annular Ring for (external, Via top) for button Plating

The button pad protrusion out of pattern pad is allow. The gap between pattern edge and next trace edge shall be more than 80% of ODB++ distance.
The minimum annular ring (button pad to via) shall be more than 12.5μm. Hole to pad breakout is not allowed.



Case	Acceptance criteria
1	Accept, annular ring (button pad to pattern pad) shall be more than 0μm
2	Reject, button pad breakout from pattern pad If gap between pattern pad and trace < 80% of ODB ++
3	Reject, annular ring (Hole to button pad) < 12.5um is not allowed.
4	Reject, Hole to button pad breakout is not allowed.

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3.4.HDI Stack-up Requirements

FPC that require the use of high density interconnect (HDI) technology must meet the requirements of the multilayer stack-up detail in the drill drawing layer of the ODB++.

HDI FPCB fabricated for Apple must meet the capture and target pad dimension and location requirements, blind and buried via dimension and location requirements, as well as through hole via dimension and location requirements as specified in the ODB++ data.
The FPC fabrication supplier will not use acrylic adhesive that will be exposed in the hole wall of blind via without approval in writing by an authorized Apple representative prior to FPCB fabrication by the supplier.

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4. Copper Plating

4.1.Copper Plating Quality and Integrity

Regardless of the plating type, all plated holes, exposed pad areas, as well as any other areas designated for plating must demonstrate complete coverage with no evidence of separation, cracks, voids or exposed underlying metal or material.
Nodules or rough plating in plated through hole shall not reduce the hole diameter below the minimum limit defined in the ODB++ file or MCO .

4.2.Plated Copper purity

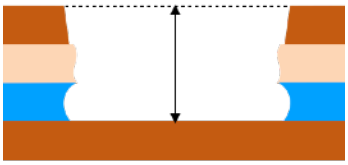
Minimum purity for plated copper on Apple FPCs shall be 99.5% as determined by IPC-TM-650 2.3.15.

4.3.Plated Copper Thickness

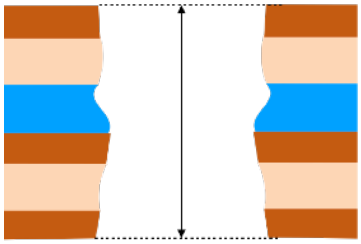
Absolute minimum plated copper thickness shall be accordance with below table, referring to **Finished Goods level thickness**, application to both **button and panel copper plating**.

Copper plating type	Hole Height/ Depth	Absolute minimum thickness of plated copper			Dimple and Protrusion
		Surface thickness	Corner thickness from top base Cu	Everywhere else (including hole wall, bottom thicknesses)	
Full filled via	All	MCO or ODB++	5 μm	Not Applicable	Protrusion ≤ 3 μm Dimple ≤ 15 μm
Non full filled via	> 300 μm	MCO or ODB++	12.5 μm	12.5 μm	Not Applicable
	≤ 300 µm ≥ 200 µm	MCO or ODB++	10.5 μm	10.5 μm	
	< 200 μm	MCO or ODB++	5 μm	7.5 μm	

- FPCB fabrication supplier shall follow QCP 080-03022 for requirements about in-process control to guaranty these thicknesses (e.g. sample size, etc.).
- Supplier shall select and define the plating type unless specified in MCO or ODB ++
- Hole height / depth refers to drill depth from layer to layer during copper plating process. Refer to below illustrations.



Hole height/depth for BVH



Hole height/depth for PTH

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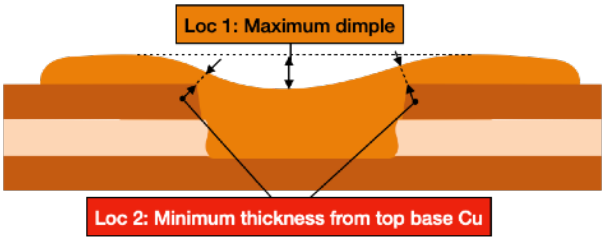
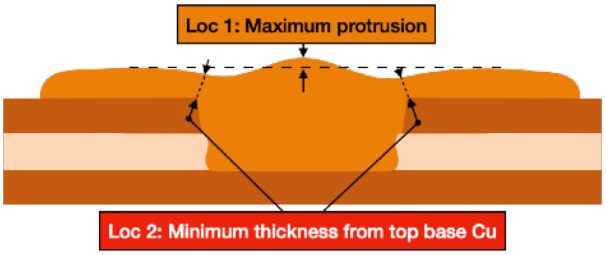
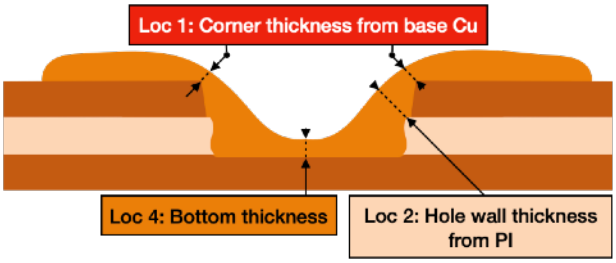
- FPCB fabrication supplier shall measure plated copper thickness at below locations depending on type of plating. Refer to below illustrations.
  - **Full filled via**:
    1. Maximum dimple or protrusion
    2. Minimum thickness from top base Cu, generally at the corner
  - **Non full filled via**: (including half filled via, Non filled via and thru hole plating)
    1. Corner thickness from **top** base Cu
    2. Wall thickness from PI
    3. Wall thickness from adhesive (if Bonding sheet/film is applied)
    4. Bottom thickness from **bottom** base Cu for BVH



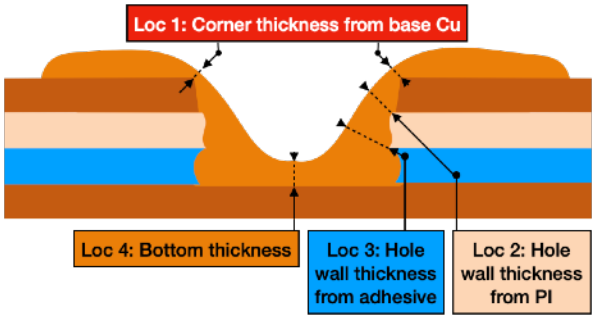
Measurement locations on BVH full filled via with dimple



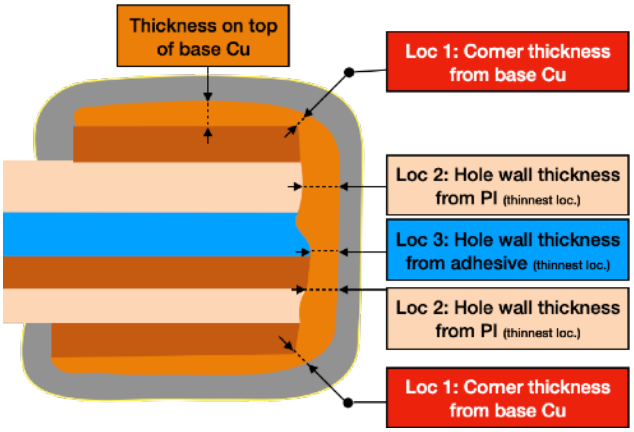
Measurement locations on BVH full filled via with protrusion



Measurement locations on BVH without adhesive



Measurement locations on BVH with adhesive



Measurement locations on PTH

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5. Conductive Patterns

5.1. Pattern Location

The maximum allowable deviation of any feature from the nominal ODB++ data location shall accordance with table 5.1 .

Table 5.1. Pattern location tolerance

Pattern Feature	Tolerance
Pattern to Pattern	±50 μm
Pattern to Outline / NPTH (except ziff pattern)	±150 μm
Pattern (Outer layer) to PTH	±75 μm

5.2. Width

The finished width of any conductor shall be measured at the **base (foot)** of the conductor. The maximum allowable deviation from the nominal width in the ODB++ data shall be 25.4 μm or 20% whichever is less.

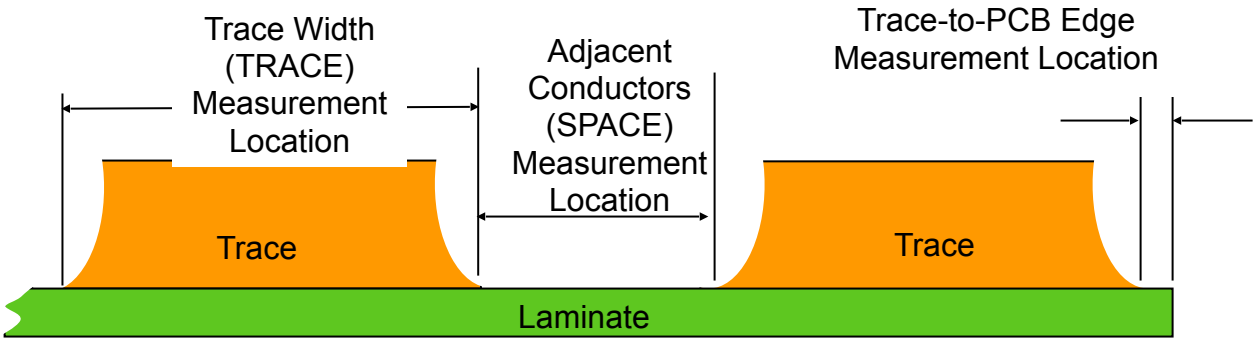

Figure 5.2.1. Trace Measurement Locations

Any trace width adjustment, which is made in order to match impedance, which conforms to the specification above, require written approval by an Apple representative

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Any Apple approved modified trace width shall be used as the nominal basis for the conductor width tolerance however, in no case may the minimum spacing between adjacent conductors be reduced below 80% of the space in the original ODB++ data, whichever is less.

The finished dimension of an exposed metal defined pad shall not deviate more than 25.4 μm or 20% whichever is less, from the dimension specified in the Apple ODB++ data.

For subtractive etch processing, the conductor width as measured at the narrowest point of the conductor (undercut) shall be no more than 13um less than conductor width at the widest point on any side for Cu thickness < 40um.
For Cu thickness between 40 to 70um, < 25um is permitted.

Table 5.2. Distance between trace foot and narrowest edge

Copper thickness	Distance between trace foot and the narrowest edge of trace
< 40 μm	< 13 μm
40 ~ 70 μm	< 25 μm
> 70 μm	Consult with SQE

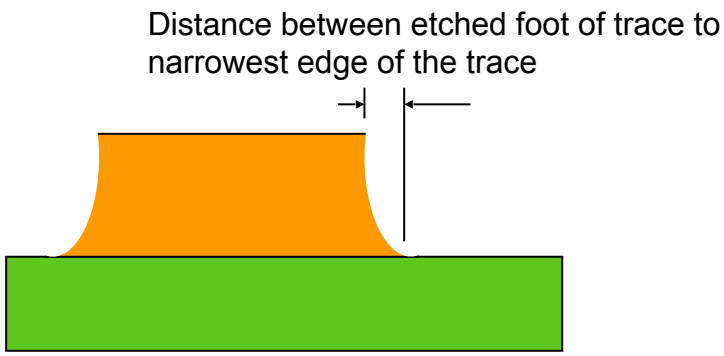

Figure 5.2.2. Exposed Metal Pad Measurement Locations

5.3.Impedance Control

Impedance Control requirements specified by Apple shall be stated in the notes below the stack up detail (unless otherwise specified) within the Drill Drawing document provided in the ODB++ data package. If no impedance tolerance is provided, it shall be ±10% for any single-ended or differential impedance traces. Impedance testing shall be

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performed on a test coupon and or using FPC. FPC fabrication supplier shall confirm impedance correlation between product and coupon.
See IPC-2251 for the equations for impedance calculation and the test method to be using TDR.

Any impedance trace width adjustment must be approved in writing by Apple EE DRI.

5.4.Repairs

Repair of outer layer and inner layer circuit traces is not allowed on Apple bare FPC and rework or repair of power or ground layers is not allowed on Apple FPC.

5.5.Separation

No portion of a conductive pattern shall exhibit evidence of separation from, or lack of adhesion to the base laminate material.

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5.6. Trace Tapers/Tear Drops

Trace tapers may not exceed the diameter of via pad and must only extend a distance 2 time equal to the via pad diameter from the center of via pad to the conductor interface before the taper must meet the width dimensions of the affected conductor. See Figure 5.6.1.

The addition of trace tapers may not violate any Apple specified requirements and the fabrication supplier must perform a design rule check after trace tapers are added to confirm no violation.

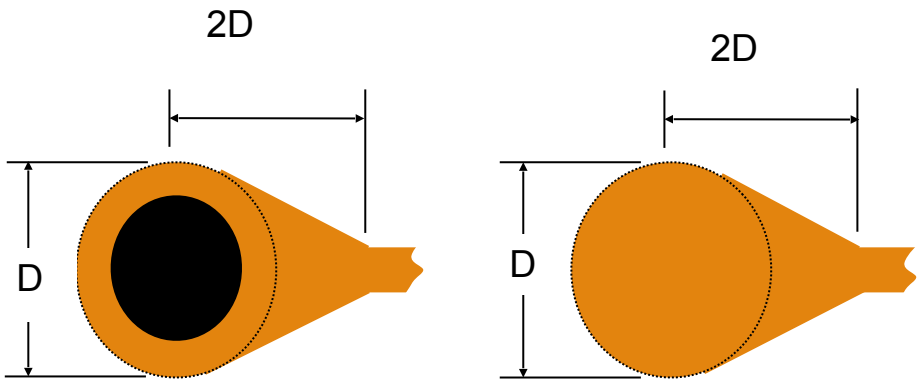

Figure 5.6.1. Trace Taper / Tear Drop modification

Tear drop shall be added to the pad/trace interface for all traces and pads including square and rectangular pads, the “D” dimension shown below may equal the longest edge of the pad or It shall capture by cover layer or photo imageable cover layer by minimum of 50 um. See Figure 5.6.2

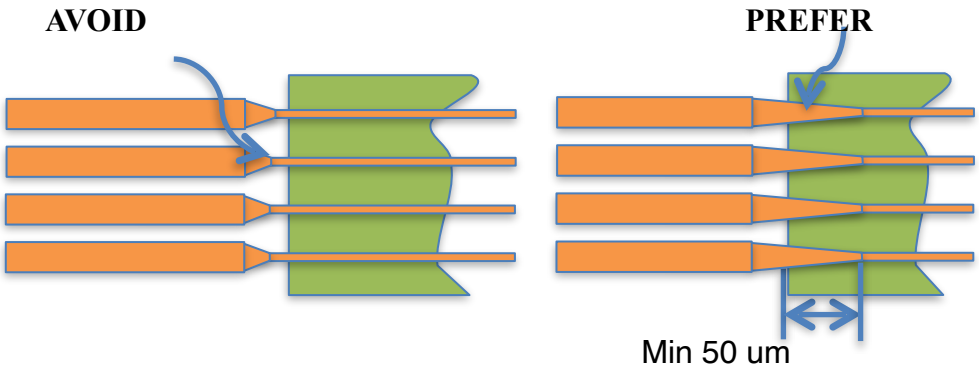

Figure 5.6.2. Trace taper / Tear drop modification (Cont.)

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6. Photo-Imageable Soldermask and Coverlay

6.1. Materials and Application

Soldermask utilised for Apple FPC may be liquid photo imageable and Apple personal approved photo imageable dry film (must conform to specific thickness requirements) must be applied over bare copper only.

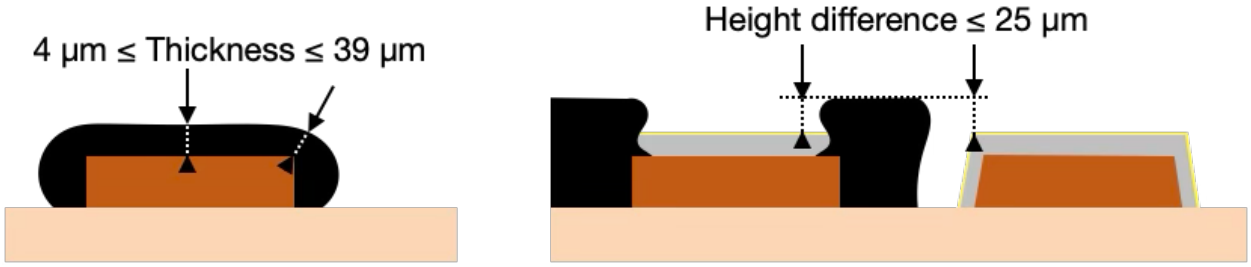
All polymeric soldermask, coverlays must be in compliance IPC 6012 or IPC 6013 as applicable when tested according to IPC-TM-650 2.6.3.1 Class H, Moisture and Insulation Resistance 100Mohm.

6.2. Photo Imageable Soldermask Thickness

Soldermask thickness shall be an absolute minimum of 4 μm at a pattern edge and an absolute maximum of 39 μm over all conductive surfaces is required by the soldermask layer of the ODB++ data.

The height difference between an exposed metallized area and the upper surface of the soldermask immediately adjacent to the metallized area may not exceed 25 μm.

Image for solder mask thickness spec



6.3. Photo Imageable Soldermask Registration

The finished size of an exposed photo imageable solder mask defined metal pad shall not deviate more than 25 μm from the size specified in the Apple ODB++ data.
The maximum soldermask registration tolerance for any FPCB is ±50 μm from the same layer copper pattern.

All metallized areas, including for example the etched foot of a feature, which are shown to be covered with solder mask shall not be exposed and shall be covered by solder mask as specified in 6.4

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At no time shall any photo imageable soldermask be allowed on any exposed metallized surface where shown to be free of soldermask on the soldermask layer of the ODB++ data.

The finished dimension of an exposed photo imageable soldermask defined metal pad shall not deviate more than 25µm from the dimension specified in the Apple ODB++ data.

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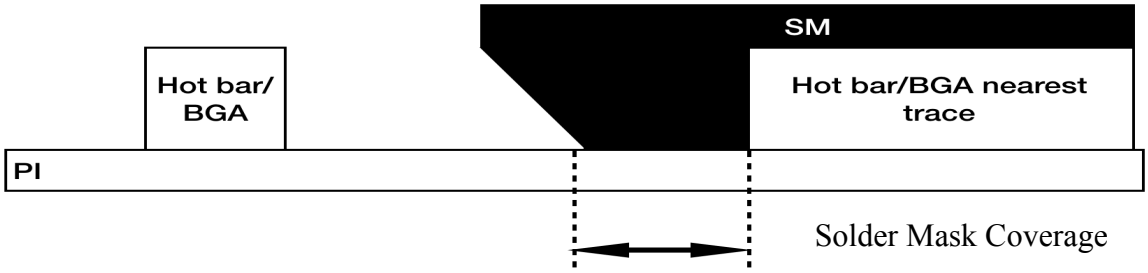
6.4.Photo Imageable Solder mask coverage

All metallized areas, including for example the etched foot of a feature, which are shown to be covered with solder mask shall not be exposed and shall be covered by solder mask

The Conductor trace exposure by solder mask is not allowed. Solder mask coverage requirement shall refer to below table.

Pattern Feature	Solder Mask Coverage
BGA / flip chip pad area / hot bar / conductor trace	Min 5μm



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6.5.Coverlay / LPI Registration opening size and Adhesive Squeeze out

At no time shall any photo imageable solder mask or coverlay be allowed on any exposed metallized surface where shown to be free of solder mask on the solder mask layer of the ODB++ data.

When copper pads are coverlay defined the coverlay shall over-lap onto the exposed copper pad by 150μm nominal and the adhesive squeeze out shall be limited to 100μm + 0 μm – 100μm. The coverlay opening in ODB++ data are already compensated for the adhesive squeeze out factor .The coverlay defined pad may not vary in size from ODB++ data by ± 50 μm.The FPCB fabrication supplier may decrease the size of the punched coverlay opening as long as the adhesive squeeze out will be controlled to the same magnitude. See Figure 6.5.1/2

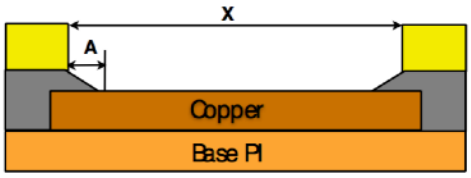

Figure 6.5.1. Coverlay opening size and Adhesive Squeeze out Allowances (Coverlay defined)

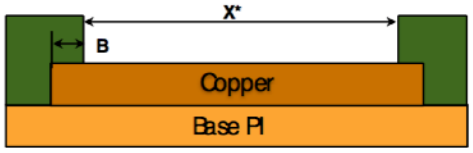

Figure 6.5.2. Solder Mask opening size (solder mask defined)

X	Coverlay opening size	± 50μm of ODB++ data
X*	Solder Mask opening size	±25 μm of ODB++ data
A	Adhesive squeeze out	Max 100 μm
B	Solder Mask Overlap	Min 0 μm

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When pads are copper defined copper shall be free of adhesive squeeze out or solder mask. The coverlay opening in ODB++ data are already compensated for the adhesive squeeze out factor. The copper define pad may not vary in size from the ODB++ data by ±25μm. See Figure 6.5.3/4

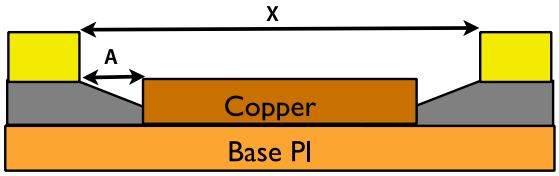

Figure 6.5.3. Coverlay Opening and Adhesive Squeeze out Allowances (copper defined)

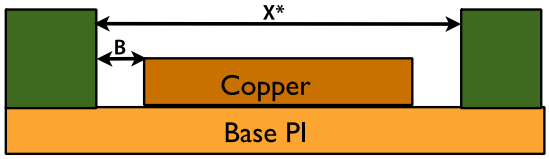

Figure 6.5.4. Solder mask Opening and offset allowances (copper defined)

X	Coverlay opening size	± 50μm of ODB++ data
X*	Solder Mask opening size	±25 μm of ODB++ data
A	Adhesive squeeze out	Clean from copper
B	Solder Mask offset	Clean from copper

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6.6.Photo Imageable Solder mask and Coverlay Overlap

Solder mask and coverlay overlap dimensions should follow Apple ODB++ and MCO. In case there are no requirements in ODB++ and MCO, FPCB shall maintain an absolute minimum of 0.25 mm overlap between the coverlay and solder mask at the transition point.

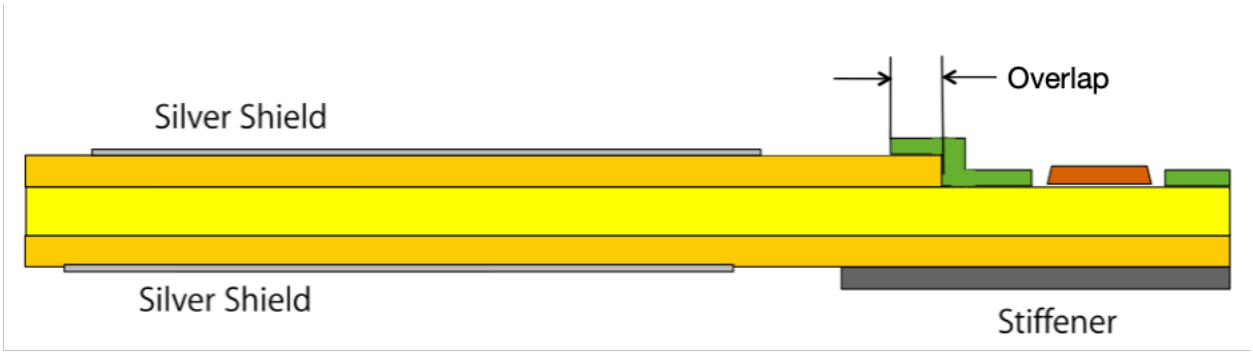

Figure 6.6. Coverlay and Solder mask overlapping requirement

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7. Surface finishing / Coating

7.1.Material thickness and Application type

Overall surface finishing or selective surface finishing requirement shall be specific in surface finishing layer of ODB++ or MCO. Reference thickness shall be see in this section .

7.2.Electroless Nickel/Immersion Gold

When specified on the fabrication drawing, all exposed areas of the entire board shall be plated with gold over nickel over base copper. ENIG with flexible Nickel is require on all Apple FPCBs where ENIG is specified as surface finish.
Immersion plated gold shall have an average thickness between 0.025 μm (1μ") thick and 0. 075 μm (3μ") thick with an absolute minimum of 0.025 μm (1μ") thick and an absolute maximum of 0.100 μm (4μ") thick and be 99% pure.
Electroless nickel shall be a minimum thickness of 2 μm and a maximum thickness 6μm with exception to iff fingers contacts that must meet an absolute minimum thickness of 2μm and a maximum thickness 3.75 μm.

7.3.Electrolytic Gold / Nickel Plating

All FPCB edge connection contacts shall be plated with gold over nickel over base copper unless otherwise stated by the ODB++ or MCO.
Plated gold used for edge connection contacts shall conform to AMS 2422 or ASTM B488.
Plated gold shall be a minimum of 0.25 μm thick and 99% pure. The hardness of the plated gold shall be 130 to 200 Knoop.
Plated nickel shall conform to the requirements of AMS 2403.
Plated nickel used for edge connection contacts shall be a low stress deposit, with an absolute minimum thickness of 2 μm and a maximum thickness 6μm with exception to Ziff fingers contacts that must meet an absolute minimum thickness of 2μm and a maximum thickness 3.75 μm.

7.4.Organic Solderability Preservative (OSP)

When identified as the surface finish for an Apple FPCB, OSP coating over bare copper shall have a minimum thickness of 0.1 μm verified by a repeatable technique.

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7.5. Electroless Nickel/Electroless Palladium/Immersion Gold (ENEPIG)

When specified on the fabrication drawing, all exposed areas of the entire board shall be plated with gold over palladium over nickel over base copper. Gold plating shall not be applied over bare copper.

Immersion plated gold shall be 0.09 μm – 0.20 μm thick and 99% pure. Electroless palladium shall be a minimum of 0.03μm thick and a maximum of 0.10μm. Electroless nickel shall be a minimum of 4.0 μm thick and a maximum of 8.0μm

7.6. Direct Electrolytic Gold

When identified as the surface finish for an Apple FPCBs, electrolytic Gold over bare copper shall be minimum of 0.4 μm and maximum of 1μm and be 99% pure.

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8. Hot Bar Pad Design

8.1.Material selection

FPC with hot bar application, Dielectric base material selection shall be stand hot bar bonding temperatures 320°C for 10 sec. Clarity / Transparency of PI material must meet section 8.6 specification. Any deviation requires approval from Apple authorized person.

8.2.Dimension specification

Specification limits to be applied to all pads within the hot bar footprint area.

Fabrication suppliers must be able to demonstrate Cpk > 1.33 for items at each hot bar footprint area present in the MCO level design. The mean value may be based on the 95% confidence limits from a sample for sample sizes less than 32pcs.

Term and Definition use in this section :

ASRO : Section A Solder Resist Opening

BSRO : Section B Solder Resist Opening

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8.2.1.Pad Dimension Specification for Panel Plated or Unplated Pads, Metal and Solder Mask Defined

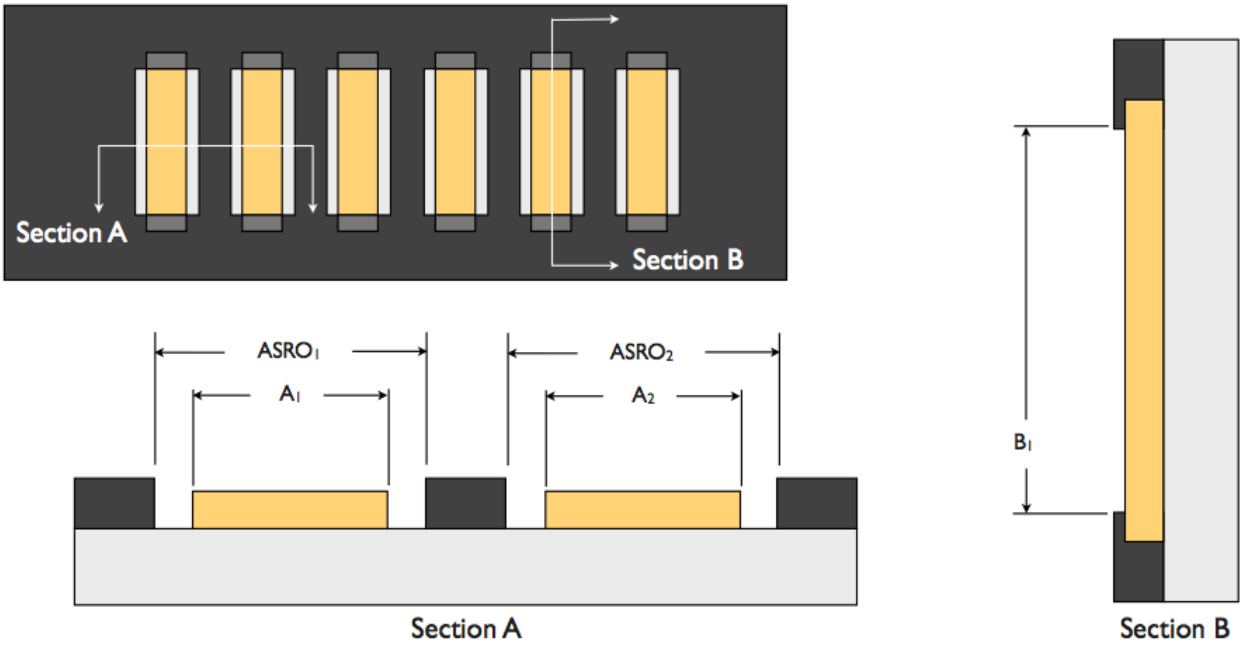

Figure 8.2.1 Expose copper or solder mask opening measurement (Metal and Mask defined)

A & B : The finished pad dimension as measured at the bottom of the pad must be within 0.025mm of the nominal conductor pad dimension shown in the MCO level design.

ASRO : The finished solder resist opening must be within 0.025 mm of the nominal opening shown in the MCO. The solder resist may not encroach on the pad when the nominal clearance from solder resist is 0.050 mm or greater.

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8.2.2.Pad Dimension Specification for Panel Plated or Unplated Pads, Metal Defined

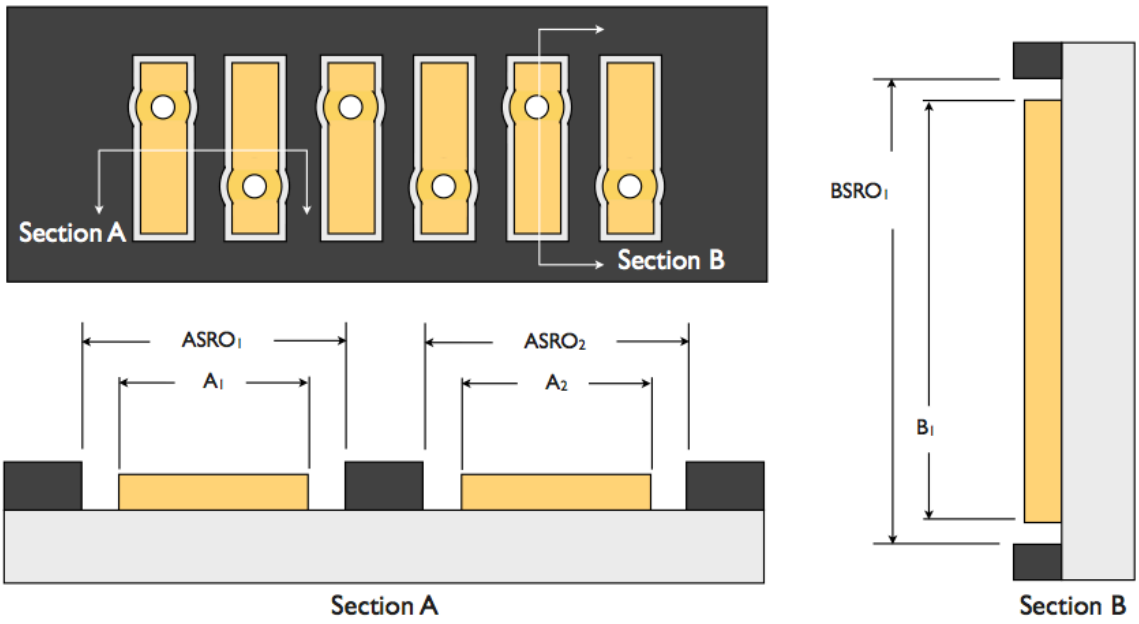

Figure 8.2.2. Expose copper or solder mask opening measurement (Metal defined)

A & B : The finished pad dimension as measured at the bottom of the pad must be within 0.025mm of the nominal conductor pad dimension shown in the MCO.

ASRO & BSRO : The finished solder resist opening must be within 0.025 mm of the nominal opening shown in the MCO. The solder resist may not encroach on the pad when the nominal clearance from solder resist is 0.050mm or greater.

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8.2.3.Pad Dimension Specification for Area Plated Pads, Metal Defined

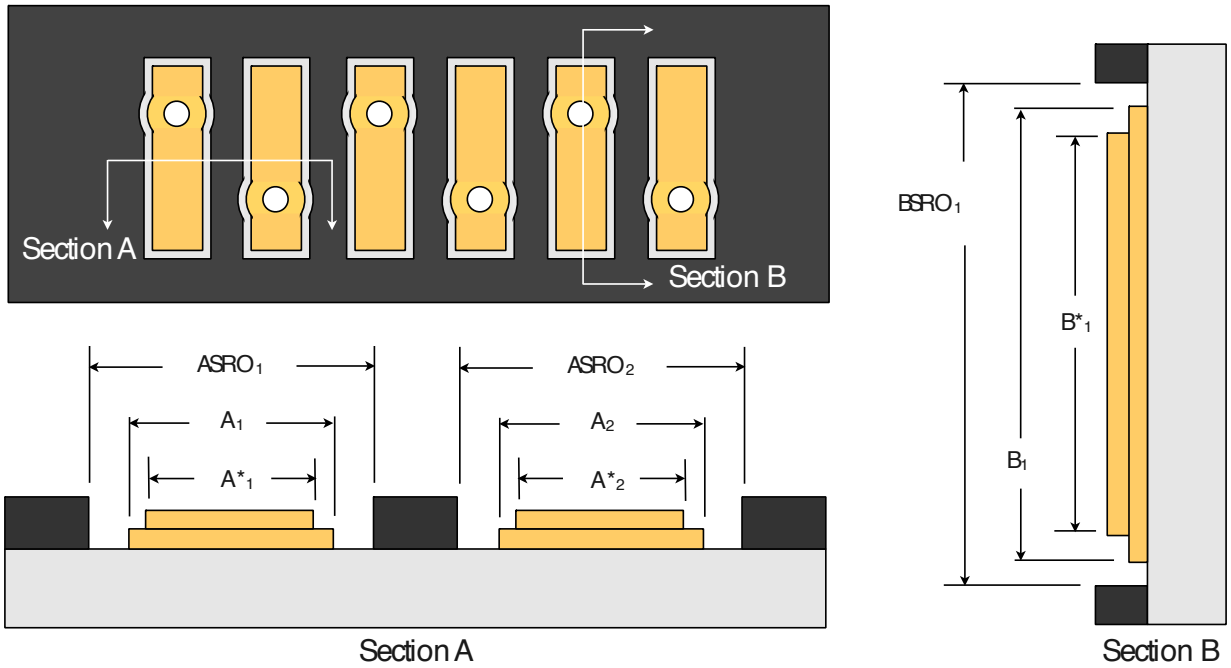

Figure 8.2.3. Expose copper or solder mask opening measurement (Metal defined, Area plated)

A & B : The finished pad dimension as measured at the bottom of the unplated pad must be within 0.025mm of the nominal conductor pad dimension shown in the MCO.

A* & B* : The finished dimension of the plated portion may not be more than 0.100mm smaller than the dimension of the unplated area.

ASRO & BSRO : The finished solder resist opening must be within 0.025mm of the nominal opening shown in the MCO. The solder resist may not encroach on the pad when the nominal clearance from solder resist is 0.050mm or greater.

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8.3.Hot Bar Pad Solder Mask Thickness

Specification limits to be applied to all pads within the hot bar footprint area.

8.3.1.Solder Mask Thickness for Panel Plated or Unplated Pads

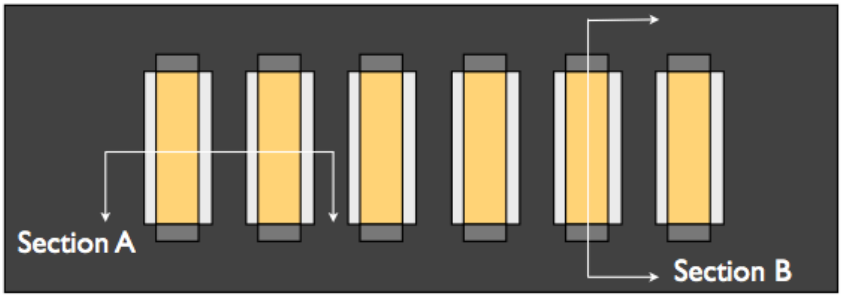
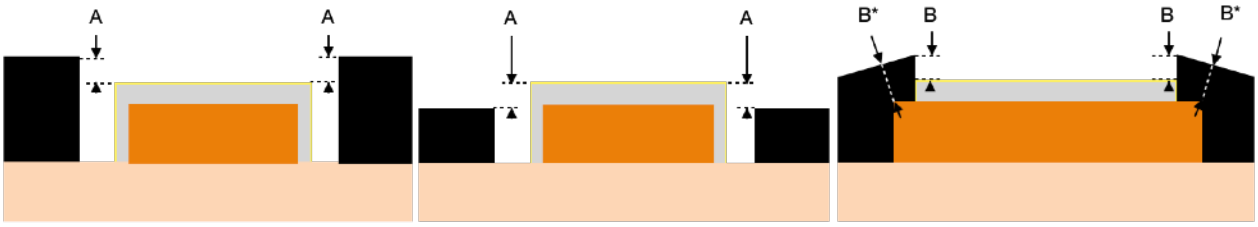


Figure 8.3.1. Example of hot bar pads (panel plated)
Section A follows NSMD pads requirements
Section B follows SMD pads requirements



Solder mask is **above** NSMD pads
A < 0.010 mm

Solder mask is **below** NSMD pads
A < 0.010 mm

Solder mask thickness measurement locations on SMD pads
0 < B < 0.020 mm
0.004 mm < B* < 0.020 mm

- **NSMD pads**
  - The mean value of solder mask height **(A)** above the exposed metal surface must be less than 0.002 mm
  - The absolute maximum height **(A)** of solder mask **above and below** the exposed metal surface may not exceed 0.010 mm.
  - This shall not apply to the specific locations with a trace between pads.
- **SMD pads**
  - The solder mask height **(B)** may not exceed 0.020 mm above the exposed metal surface.
  - Solder mask thickness **(B*)** on the covered portion of the pad, including pad's edge, must be more than 0.004 mm.

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8.3.2.Solder Mask Thickness for Area with Button Plated Pads

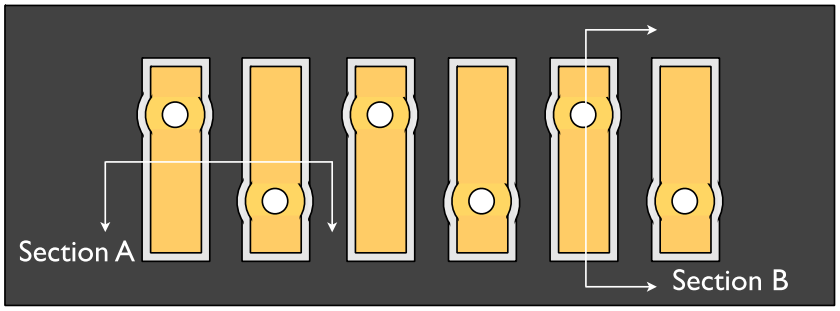
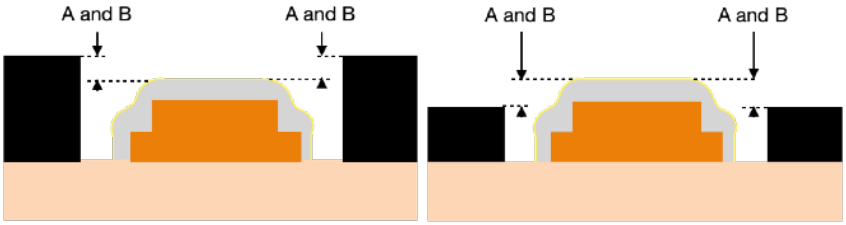


Figure 8.3.2. Example of hot bar pads (button plated)
Section A and B follow NSMD pads requirements



Solder mask is **above** NSMD pads
A and B < 0.010 mm

Solder mask is **below** NSMD pads
A and B < 0.010 mm

- **NSMD pads**
  - The mean value of solder mask height **(A & B) above** the exposed metal surface must be less than 0.002 mm.
  - The absolute maximum height **(A & B)** of solder mask **above and below** the exposed metal surface may not exceed 0.010 mm.
  - This shall not apply to the specific locations with a trace between pads.

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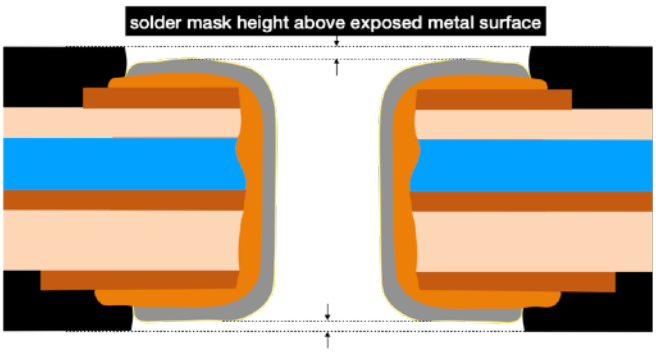
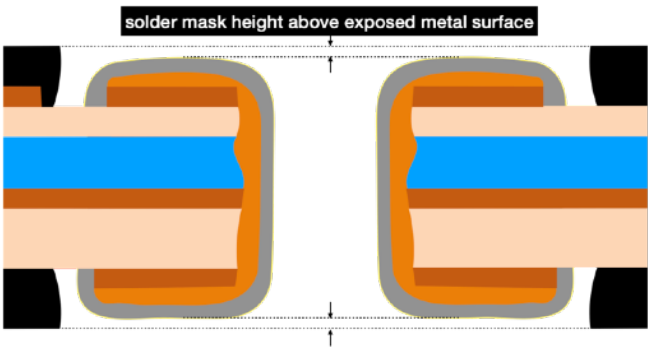
8.3.3.Solder Mask Thickness for Watch Programs

FPCB fabrication supplier to seek Apple DFM DRI approval in written before implementing LPI thickness controls on hot bar flexes.
Unless specify in MCO or Apple program DRI, the solder mask height **above** the exposed metal surface must be more than 0.000 mm and the absolute maximum height of solder mask above the exposed metal surface may not exceed 0.015 mm.

Refer to the below images for visuals.
Solder mask height = 0 ~ 15 μm (Solder mask surface always **above** metal surface)



Measurement locations on SMD pads



Measurement locations on NSMD pads

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8.4.Pre-soldered Thickness (After Reflow)

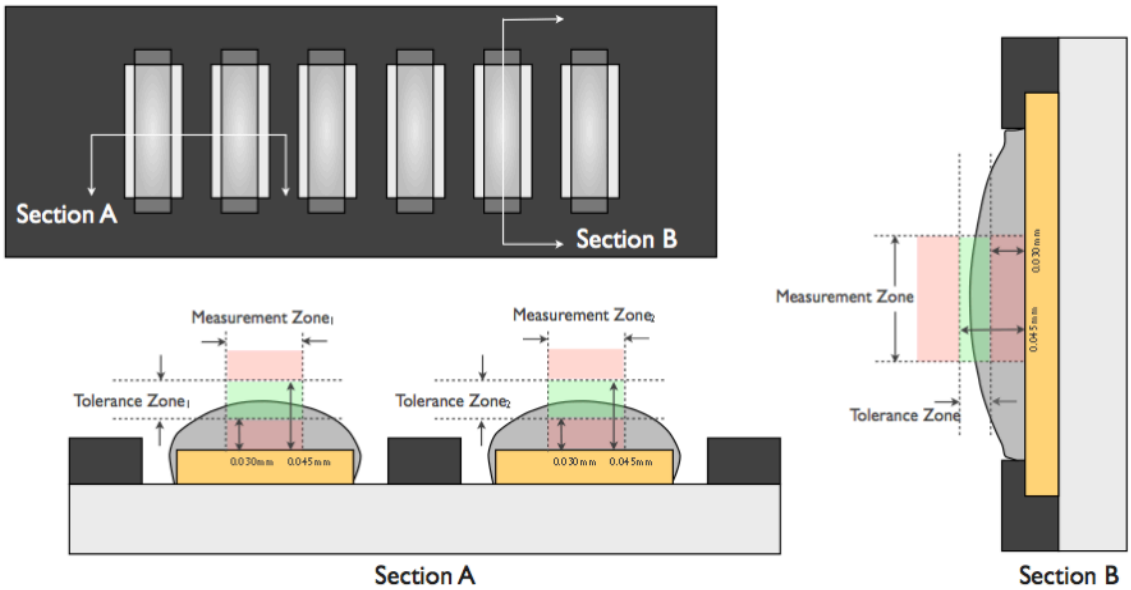

Figure 8.4. Pre-Solder Thickness (after reflow)

The Measurement Zone is defined as the middle 50% of the exposed pad width.
The Tolerance Zone is defined as the area 0.030mm to 0.060mm above the exposed pad.
Within the Measurement Zone, the height of solder after reflow must be completely within the Tolerance Zone.

Cross-section shall be the official method of measurement and vendors must be able to show Cpk > 1.33.
Vendors may propose a non-destructive method of solder height verification. An acceptable method must demonstrate correlation to cross-section measurements with $R^2 > 0.90$ for values within the Tolerance Zone.

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8.5.Dimensional Stability and pre conditioning test

FPCBs with Hot bar application shall be free of winkle, debris at hot bar pads area and maintain surface flatness**.** When measured at fiducial center to center under normal lay out condition or flatten lay out condition the distance must be within ±0.025mm from ODB++ data. See Figure 8.5

At the preconditioning test of 3time, 320°C, 5 sec solder bath or hot oil dip the solder mask blistering, burning, delamination or color change shall meet IPC 6013 delamination/ blister specification .

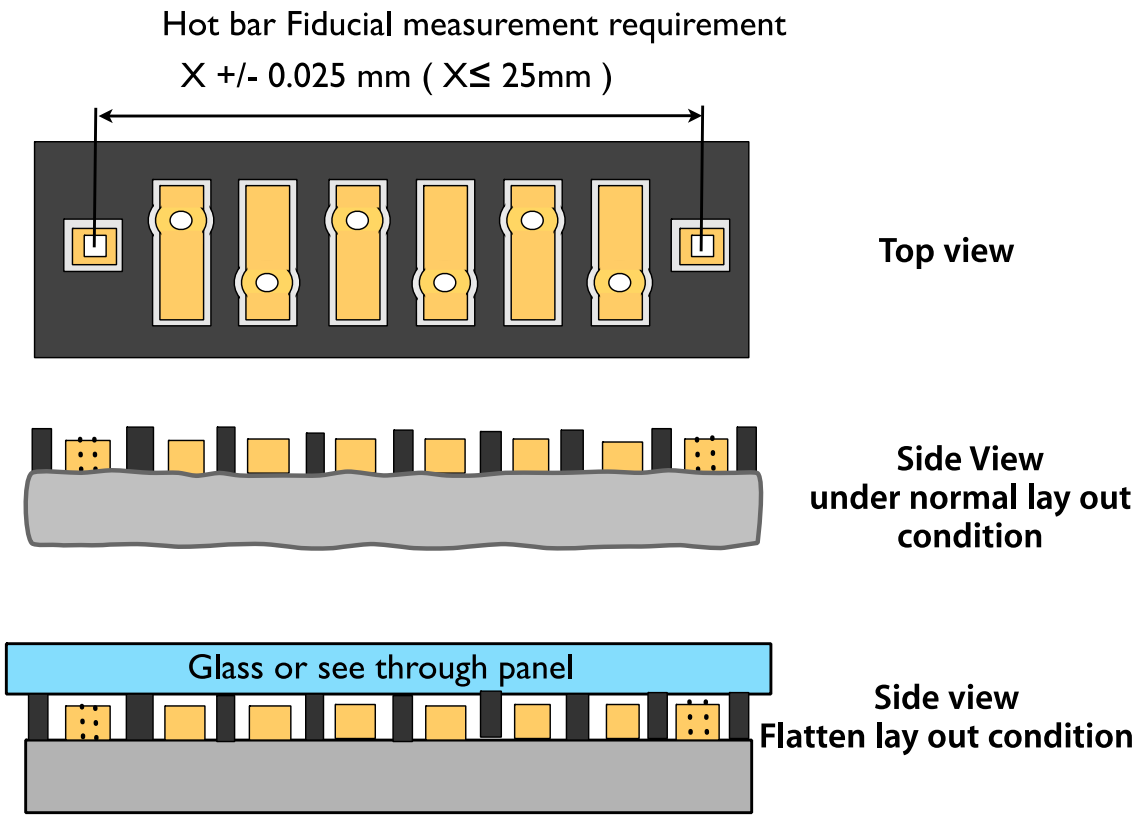

Figure 8.5. Hot bar fiducial measurement condition

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8.6.Fiducial Transparency

For FPC with hot bar application, FPCB fabrication supplier must perform transparency test on the material used for fabrication according to ASTM D-1003 using BYK Gardner, Haze-Guard Plus or equivalent equipment .

When measured from base material, fiducial at the hot bar bonding area shall meet transparency data >60% and Haze data <80%.

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9. Mechanical and Dimensional Requirements

9.1.FPCB Thickness

The finished total thickness of the FPCB shall be measured after the completion of all fabrication operations required for the specific design to match the ODB++ data and shall be measured from the solder mask/coverlay surface over conductors of one side of the FPCB to solder mask/coverlay over conductors on the other side. Section with different layer stack up required separate measurement for each section. See Figure 9.1

Stackup Structure			Layer Descriptions	Raw Material thickness	The theoretical thickness (After compressed)			Raw-material Suppliers	Product code
Zone A	Zone B	Zone C			Zone A	Zone B	Zone C		
			SUS Stiffener	100	100			XXXXXX	XXXXXX
			Stiffener Adhesive	60	40			XXXXXX	XXXXXX
			Coverlay Polyimide	12	12	12		XXXXXX	XXXXXX
			Adhesive (modified epoxy)	20	15	15			
			13um Button Plating	13	-	-	-	XXXXXX	XXXXXX
			Base ED Cu	12	10	10		XXXXXX	XXXXXX
			PI Base Film (Adhesiveless SS)	12	12	12			
			Bonding Sheet	20	15	15		XXXXXX	XXXXXX
			Coverlay Polyimide	12	12	12		XXXXXX	XXXXXX
			Adhesive (modified epoxy)	20	15	15			
			Base ED Cu	12	10	10	10	XXXXXX	XXXXXX
			PI Base Film (Adhesiveless DS)	25	25	25	25		
			Base ED Cu	12	10	10			
			13um Button Plating	13	-	-	-	XXXXXX	XXXXXX
			Adhesive (modified epoxy)	20		15		XXXXXX	XXXXXX
			Coverlay Polyimide	12		12			
			LPI	20	20			XXXXXX	XXXXXX
			Total After Press Thickness (um)		296	163	35		

Figure 9.1 Prefer Material Stack up Sample

An FPC with gold plated edge connectors (sometimes referred to as gold fingers or gold tabs) may have a separate thickness requirement measured over the gold plated connector area and should be verified in addition to the solder mask/coverlay to solder mask/coverlay thickness requirement for the rest of the FPCB.

FPCB thickness tolerance shall be a maximum of ±10% of the nominal FPCB thickness unless otherwise specified on the drill-drawing layer of the ODB++ data or the MCO.

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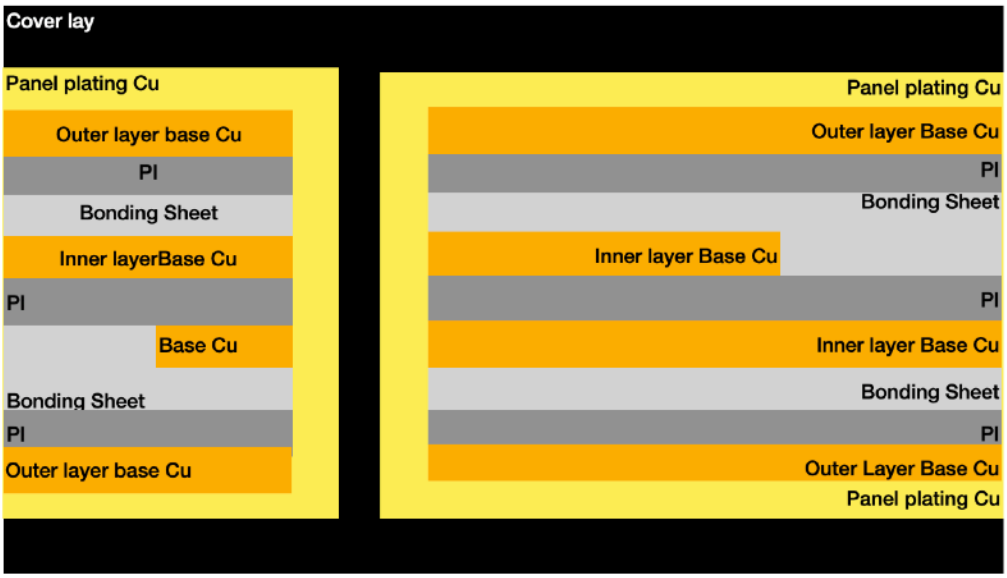
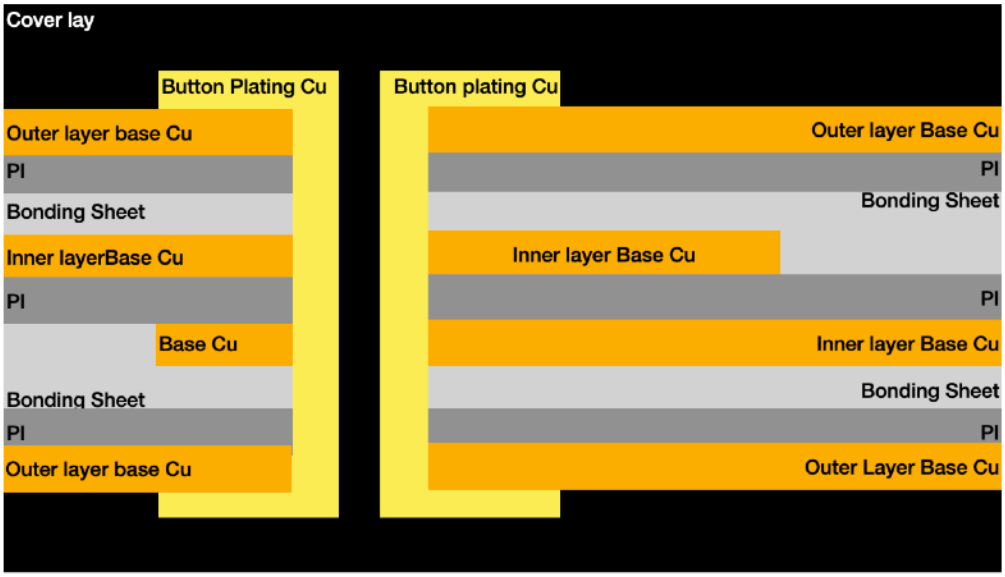
An FPC with panel plated Cu may have a separate thickness requirement measured over the base Cu with refer to MCO requirement

Below picture show the example of stack up with panel plating Cu and Button Plating

Panel Plating



Button Plating



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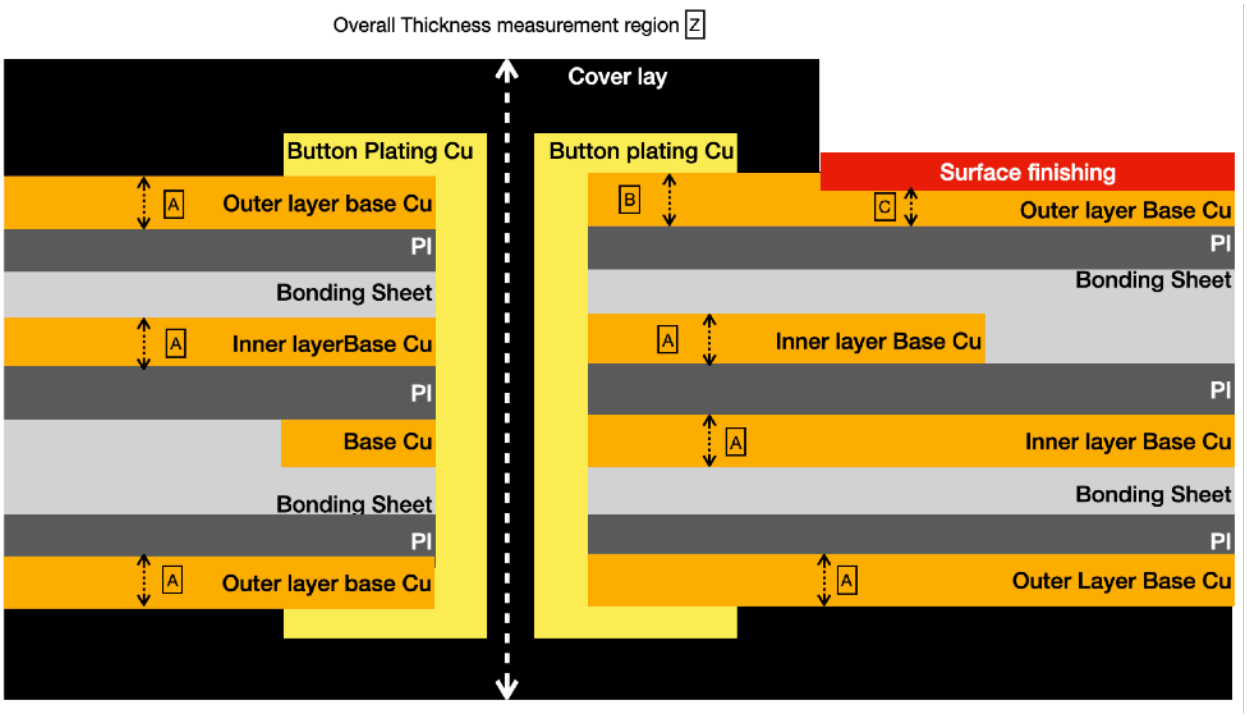
With refer to MCO requirement on the overall stackup thickness, the overall thickness measurement region shall measure at the button plating area. Measurement shall be taken as shown on below picture [Z] The outer layer can be refer to Coverlay to Coverlay or Solder Mask to solder mask or Coverlay to solder mask which define by MCO



An FPC with button plated Cu, the MCO specified Cu thickness should be measure base Cu [A point] and avoid to measure underneath the button plated Cu area [B point] and underneath ENIG area [C point]

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9.2.FPCB Outline

The physical features and dimensions of the FPC outline shall be specified in the drill drawing layer of the ODB++ data and the MCO. The FPC shall be punched as specified in the ODB++ data for singulation purposes. See Figure 9.2

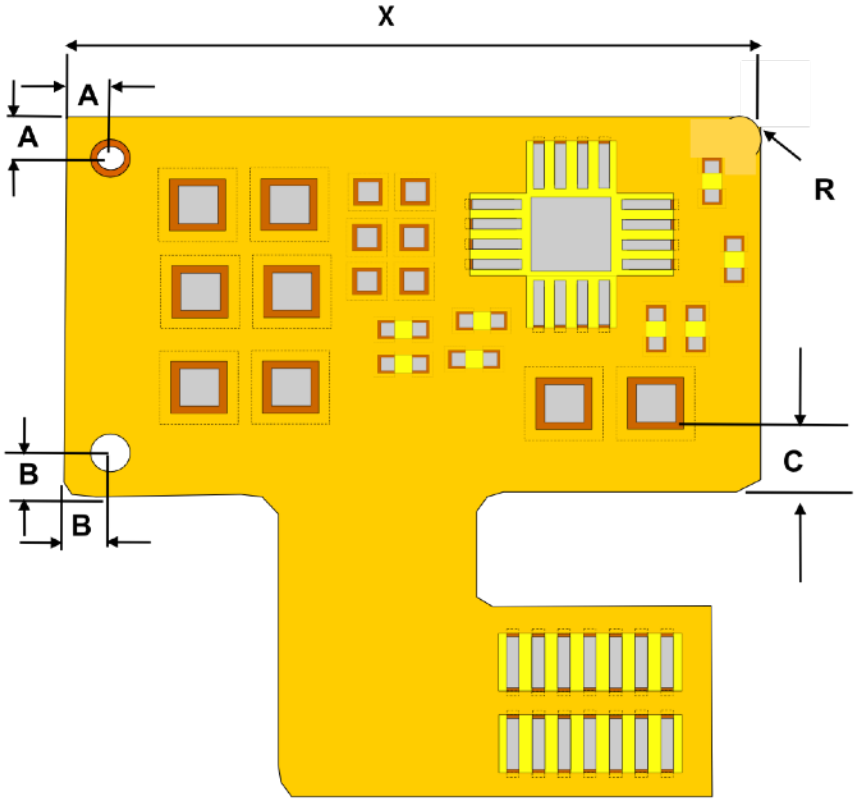

Figure 9.2. Flex outline Description

Table 9.2. FPCB Outline Tolerance Requirements

X	Outline to Outline tolerance	±200 μm
A	Outline to P.T.H. tolerance	±200μm
B	Outline to N.P.T.H. tolerance	±200μm
C	Outline to pattern tolerance (Except Ziff Pattern)	±200μm
R	Minimum outline radius	0.2 mm

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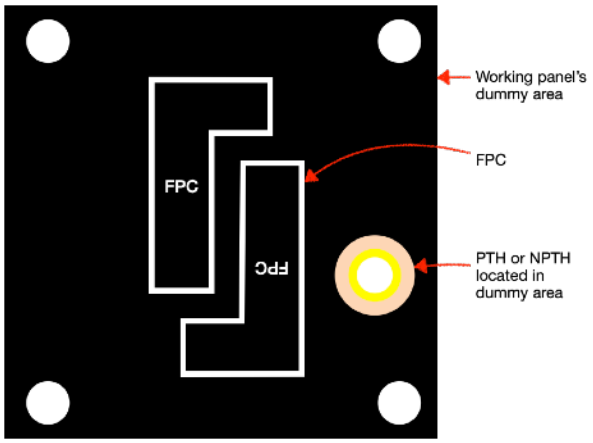
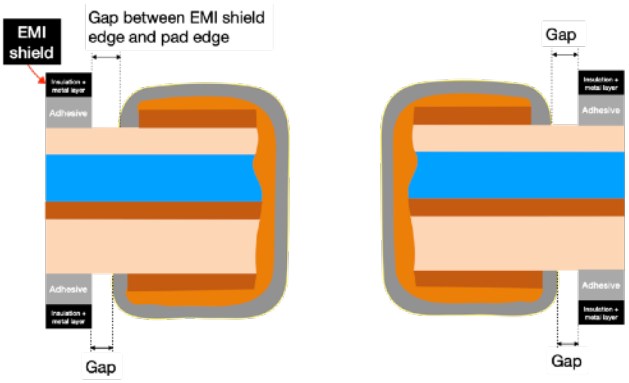
9.3. Working panel or shipping array dummy areas

FPCB fabrication supplier shall design a gap between EMI opening edge and PTH or NPTH in the dummy or unwanted areas of working panel.
The specific dimension of that gap is controlled internally by the FPCB fabrication supplier. E.g. 0.15 mm.
Refer to the below for illustrations.



Top view of working panel, detailing PTH/NPTH location in the dummy area



Cross section view of PTH in dummy area

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10.Stiffener Application

10.1.Material

Material selection and thickness requirement shall be specific in stiffener layer of ODB+ + data or MCO. Stiffener profile shall be either punch out together with outline or shall be apply individually.
SUS used for conductivity, the conductivity resistance value shall be less than 5 Ohm or as specified in MCO. SUS with conductive adhesive shall be Nickel coated with thickness range of 1μm ~ 3 μm.
The combined bow and twist of each finished FPCB at stiffener portions shall not exceed 0.75% for applications that use SMT components and 1.5% for all other applications along the linear dimension of the FPC.

10.2.Visual inspection

Refer to bare FPC cosmetic specifications 080-03267 for bare FPCB fabrication supplier OQC, and SMT/MI/CM/FATP IQC.

10.3.Stiffener Bond Strength to FPC

Refer to bare FPC Quality Control Plan 080-03022.

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11.Legend Marking

11.1.Application and Information

Vendor markings shall be present on all Apple FPCB's. Marking location shall be accordance with ODB++ data and or MCO description. Vendor markings must be located so as to be visible after assembly.

At no time shall any legend marking be applied over, or make contact with, any exposed metallized surface or at bending area zone shown to be free of legend on the silkscreen or ink jet layer of the ODB++ data unless otherwise specified.

If the FPCB design is so small that there is not enough room to apply all fabrication supplier markings, the order of precedence (of markings to apply) as space permits shall be:

1st = Fabrication supplier Logo (or equivalent mark that can identified fabrication supplier) and APN
2nd = Lot number
3rd = Sheet #
4th = Cavity number
5th = Date code
6th = UL mark/logo

Any deviation require approval from Apple authorized person.

Part numbers other than the Apple part number may not be applied to Apple FPC, unless specifically authorized in writing by an authorized person.

Country of manufacture (adjacent to the manufacturers logo) shall be marked by legend on the FPC only if specified in the ODB++ data.

The date code shall at a minimum identify the Week and Year of manufacture, and shall be applied either by legend or laser by a permanent non-conductive ink.

The UL mark applied to the FPC shall correspond to the applicable qualification file the fabrication supplier has for the FPC and shall be marked by legend.

Any other markings as required by Apple ODB++ data and/or pertinent specifications, shall be printed, or affixed to the FPCB as required.

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11.2. Legend Ink color

Legend ink color will be specified in the README file included with the data package. Table 11.1 shows the colors that must be available. Cosmetic coating specification requirement shall be same as legend marking specification unless otherwise specific in MCO.

Table 11.1. Legend Ink Colors

Development	Production
White	White

11.3. Accuracy

Reference designation, as well as all other legend marking shall be sharp and legible without magnification.
Legend marking shall be registered within 300μm of the true position specified by the ODB++ data and/or the MCO.

11.4. Durability

Legend and other non-etched markings shall consist of non-conductive inert ink capable of withstanding solvents; thermal shock, ultrasonic cleaning and all other FPCB cleaning processes related to normal FPCB/FPCBA production operations.

Legend markings must be permanent and shall no evidence of softening, smearing, or obliteration when tested with Freon TMS per IPC-TM-650.

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12.Cleanliness

12.1.Cleanliness

Cleanliness shall be in accordance with the requirements of Apple PCB Fab Process Cleanliness Requirements 080-02184, and applicable section of IPC 6013.

12.2.In-Process Cleanliness

All phases of the FPCB fabrication must be kept free from sources of contamination or degradation.
Prior to the application of soldermask and/or Coverlay, the uncoated FPCB shall meet the requirements of IPC-A-600 paragraph 19, when measured per IPC-TM-650.

12.3.Ionic Contamination

Ionic contamination for all Apple FPCB's must be less than 4.5 micrograms NaCl per square inch.
refer to Apple Bare FPC General Continuous Quality Reliability Assurance and OQC Metrology Requirement 080-02999

12.4.Wire bonding pad Cosmetic and Cleaning

To ensure Bondability and reliability of wire bonding FPCBs shall be clean , the cleaning specification shall agree upon fabrication supplier and contract manufacturer where the wire bonding process perform .
The wire bonding pad include deformation , foreign material , dent , bump , scratches and or solder paste are not acceptable and cause of rejection. Refer to 080-03267 Apple FPC incoming and outgoing Cosmetic Criteria for criteria details.

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13.Electrical Requirements

13.1.Electrical Test

Test shall be completed in accordance with all applicable IPC -TM-650 specification requirements unless otherwise specific by MCO or Apple authorized person.
Two-sided simultaneous ("clamshell") test or a flying probe which can probe both sides is required for all Apple FPCB's, which have net terminations on both sides of the FPCB.
Test programs shall be extracted from the Apple reference net list or ODB++ data.

As specified by an authorized Apple representative, 100% test coverage will be required for FPCB's used for engineering development. Report documenting the Test coverage and production lot test result shall be furnished to Apple's designated representative upon request.
If flying probe is used, all adjacent nets including those in adjacent layers must be tested for short circuit. Where the resistance requirement of under 1 ohm or trace width and spacing less than 50 µm it shall use 4 wire testing method .

13.2.Impedance Test

Unless otherwise specified on the drill drawing, test for impedance shall be performed in accordance with IPC-TM-650 2.5.5.7.
The FPCB shall be fabricated such that the impedance requirements as specified on the multilayer stack-up detail from the drill-drawing layer of the ODB++ data are properly satisfied. A test configuration, which allows for a conductive path to each applicable signal layer, which shall have trace width equivalent to the affected layer, must be provided in order to verify the impedance values. The length of the conductive path may be modified to allow for FPCB dimensions if the length and test equipment capability satisfy IPC-TM-650 2.5.5.7.

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14.Quality Requirements

14.1.Deposition

Apple will maintain the authority for final disposition of all defective FPCB's fabricated by the FPCB fabrication supplier.

14.2.Visual Inspection

Upon receipt, all FPCB's shall be subjected to visual inspection per FPC cosmetic specification 080-03267

14.3.OK2Ship Report

The FCB fabrication supplier shall furnish OK2Ship report for each shipment lot of FPCs. The inspection ands test item refer to:

- NPI sample build (from P0 to EVT included) to use OK2Ship report APN 080-02882 (NPI OK2Ship Report for Bare FPC)
- Mass Production build (after EVT) to use OK2Ship report APN 080-02881 (MP OK2Ship Report for Bare FPC)

to Apple designated EMS subcontractor as well as all designated Apple representatives.

A hardcopy shall be included in the shipping package or shall be emailed to respective Apple DRI. The FAI report shall all items specified by the MCO.

The FPCB fabrication supplier shall be responsible for any FPCB fabrication processes subcontracted to a third party supplier.

14.4.In Process Inspection

An authorized representative designated by Apple may perform, witness, or participate in such inspections and/or tests as considered necessary to ensure conformance with applicable requirements.
All drilled holes shall be inspected to ensure complete extension between the layers specified subsequent to the de-smear process.

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14.5. Reports

The FPCB fabrication supplier shall have performed and have records of capability studies for their major process in accordance with 080-01409 Apple Bare FPC Critical Process Parameters,
080-03022 Apple Bare FPC Quality Control Plan, and 080-02647 PCB FPC Line Qualification Guidelines Line, IPC-9191 or equivalent SPC program
Processes and Apple specified parameter, which are not supported by Cpk 1.33, shall be 100% inspected

Report documenting the detail of the capability studied shall be furnished to Apple's designated representative upon request

14.6. Moisture

FPCB's shall be tested for moisture content in accordance with Apple Bare FPC General Continuous Quality Reliability Assurance and OQC Metrology Requirement 080-02999
FPCB's for Hot Bar application shall maintain MSL level 5

14.7. Compliance to Industry Standards

FPCB's shall be fabricated by the FPCB fabrication supplier using best industry practices, which meet or exceed the requirements of this specification and the ODB++ data furnished by Apple.
FPCB's fabricated for Apple shall exhibit proper workmanship and there shall be no presence of residual fabrication material, corrosive fluxes, oils, foreign matter, fingerprints, bridged conductors, incomplete etching, or unauthorised rework.
Apple reserves the right to perform any inspection, test, or analysis deemed necessary to verify that each delivered FPCB conforms to the requirements as stipulated in this specification

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15.Environmental Test Methods

15.1.Continue Quality Reliability Assurance

FPCB's shall be tested for environmental (reliability Test). The test method accordance with Apple Bare FPC General Continuous Quality Reliability Assurance and OQC Metrology Requirement 080-02999

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16.Storage and Packaging

16.1.Cleanliness and Moisture

Prior to the packaging of FPC for storage and/or shipment, the FPCB fabrication supplier shall ensure that the FPC meet the cleanliness and moisture content requirements of this specification.

16.2.Storage Conditions

Storage areas for FPCB's prior to packaging shall be in a controlled environment, maintained constantly within the following specifications:

Temperature	24 ±3°C (75 ±5°F)
Relative Humidity	50% ±10%RH

16.3.Packaging

The Bare FPC The packaging method and requirement accordance with Apple Bare FPC Critical Process Parameters 080-01409

Below area need to be control and requirement accordance with Apple Bare FPC Critical Process Parameters 080-01409

- Final baking area for finish bare FPC before packaging
- Finish good bare FPC Packaging area
- Storage area for finish good bare FPC after packaging

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16.4. Shelf Life

A. FIFO system in accordance to the manufacturing date code must be applied on product shipment sequence. The maximum shelf life for Apple Bare FPC is 12 months started from FPC surface plating finishing to Assembly soldering (excluding Tin plating, Tin plating surface finishing shelf life to be specified by independent project QCP).
B. The Bare FPC finish good shall be follow the packaging and storage requirement accordance to 080-01409
C. The 12 month shelf life is referred to Bare FPC prior to component assembly and soldering. FPC with less than 4 months shelf life to assembly soldering may not be shipped to an Apple contract manufacturer without the test stated according to Bare FPC shelf life Extension 080-02864
D. Bare FPC older than 12 months to assembly may be accepted subject to test requirements according to Bare FPC shelf life Extension 080-02864
E. Overdue/Ageing Bare FPC must be subjected to the test requirements with guarantee letter approved by Apple SQE before shipment to contract manufacturer. This requirements are only applicable if the initial packaging shows no sign of tempering. If desiccant and moisture indicator card has shown sign of moisture, Bare FPC shall be scrapped. Bare FPC that pass ageing test shall be extended for another 6 months with a maximum limit of 2 years from the Surface plating finishing.
F. Guarantee letter and bundled reports of each lot must be shared with contract manufacturer SQE and Apple provided distribution list (if any) before shipment. Shipment boxes shall be clearly labeled with *ageing test done*, the date tested and the number of re-test done to date.
G. Any Bare FPC in storage manufacturers which has a shelf life > 2 year and expire / obsolete need to follow “obsolete” . “Expire” deposition matrix according to 080-4112 requirement.

16.5. Retained samples

FPCB fabrication supplier should have retaining finish good Bare FPC sample for each manufacturing lot at fabrication factory or site.

- For MP build
  - Each lot minimum 5 pieces. The finish good bare FPC sample should not be subject to any CQRA test item and be functionally good. Sample shall be kept until part EOL or 5 years whichever longer.
- for NPI build
  - All lots built leftover, all defective samples which did not ship to downstream and all the quality test samples [from CQRA, IPQC, FAI, etc] shall kept until part EOL or 5 years whichever longer.

Retained sample shall be packed and stored in the conditions as 16.2 and 16.3

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17.Shipping Requirements

17.1.Shipping Requirements

FPCB's must be shipped to Apple's designated electronic contract manufacturer within one week of the date of manufacture unless otherwise specified in writing by Apple an authorized representative.

All package must contain desiccant packets to ensure the absence of moisture, moisture colour indicator must be placed in all packages. When shipping bare FPCB's to Apple or a Contract Manufacturer the FPCB's shall be oriented in the same direction within the package. A stiffener must be placed in each package to protect the FPCB's from any unintended bending / creasing caused from handling or the crushing force of the vacuum sealing process.
The preferred weight for each package is less than or equal to 25 pounds. All packages within boxes must be labeled with the following
Traceability Markings
Date codes and lot codes
Manufacturer
Apple Part Number and Revision
Quantity of both FPCB's and Panel Arrays, if applicable.
No. of X-Outs per panel, if applicable

Warranty conditions shall be as specified on the Apple purchase order.
The FPCB fabrication supplier shall be liable for all FPCB shipments until the shipment is properly received and accepted by Apple, Apple's designated subcontract manufacturer, or other subcontractor location designated by Apple.

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Appendix A: Revision history

Rev	ECO	Release date	Released by	Description
A	382570	25 May 2005	JB	Initial release
B	612291	25 Jul 2008	DP	Revised multiple sections
C	0000903423	6 May 2010	MS/DP	Revised multiple sections including outline tolerances and plating thicknesses.
D	0001177722	27 Jun 2011	MS	Added flexibility testing requirement
E	0001926860	12 Apr 2013	PW	Revised multiple sections
F	0033286727	19 Jan 2022	W. Li L. Yuan YG Tan B. Huneau YG Zhang H. Abe	Revised multiple sections
G	0034940657	22 Apr 2022	YF Tan	1.10 Regulatory Requirements - revise the UL prepare and submission and UL approval in build stage. 4.3 Revise board thickness to hole height/depth 4.3 Added note The plated thickness shall refer to FG level thickness. 4.3 Remove Cu plating IPQC sample size and refer to QCP 6.3 Add Solder mask registration 8.3.1 and 8.3.2 Revise back revision E the Solder Mask thickness in the description and table
H	0044150591	16 May 2023	B. Huneau	• **1.7. Responsibilities**: added requirements about average and process target • **1.10. Regulatory requirements:** clarified UL certification requirements. • **2.5. Surface imperfections**: removed typos • **3.2. Hole diameter**: added illustrations and removed overlaps • **4.3. Plated copper thickness**: clarified requirements, added illustrations • **6.6. Solder mask and coverlay overlap**: updated spec • **8.3. Hot bar solder mask thickness**: corrected spec and illustrations • **9.3. Working panel or shipping array dummy areas**: added requirements • **16.5. Retained samples**: clarified requirements • Standardized naming. Replaced “FPC vendor” by “FPCB fabrication supplier” for consistency

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Rev	ECO	Release date	Released by	Description
J	0061233406	29 Nov 2024	YF Tan	• Remove document 080-03269 Apple bare FPC Outgoing Cosmetic Criteria and replace to 080-03267 Apple bare FPC Incoming Cosmetic Criteria in pages 7,14,49,52,54. • Corrected spelling, Figure number and table number • 16.5. added NPI retail sample requirement • 5.2 include Cu thickness between 40 to 70um, < 25um is permitted • 8.3.2 add drawings for explanation Solder Mask Thickness for Area with Button Plated Pad • **1.7. Responsibilities**: clarified requirement • **6.2. Photo Imageable Soldermask Thickness**: added image

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